The Geometric Engine of Time: A Proposal for a Standardization of the Cubit

BioChain AI — SHD-CCP Architecture Research

Consolidated Technical Report (Continued Series)

Abstract

This report formalizes the local acoustic cubit as a continued installment of the SHD-CCP research series and an addition to The Geometric Engine of Time. Defined as one-sixteenth of an acoustic wavelength at a fixed calibration frequency, $C_0 = v/(16\, f_{cal})$, the cubit is realized at 20 °C with an expanded uncertainty of 0.22 mm ($k = 2$) from the Cramer (1993) real-air equation, and it grows by 0.98 mm per kelvin. Every number and identity of the October 2026 technical report was recomputed in code: all seven of its mathematical results hold once stated precisely, five after correction, while its pressure coefficient is at least 55 times too large. The stellar, galactic and holonomy extensions lie at least five orders of magnitude below the realization uncertainty and cannot be tested on Earth, and the 53.22 cm cosmological cubit carries no evidential weight under a look-elsewhere test. Against climate data, calibration at the annual mean temperature fits historical cubits no better than chance; a cold-season variant at two-thirds of a 432 Hz wavelength fits the Egyptian and Nippur cubits and is preregistered as hypothesis H1b. The Geometric Engine of Time supplies the most accurate method at four points: exact caesium-cycle bookkeeping, the Triple Quad Formula (TQF) identity in rational arithmetic, a 30-node wheel that tests 20 % fewer prime candidates than the base-6 wheel, and a 0.5 s closure window that holds whole periods of the caesium clock and of both series carriers. Three preregistered protocols state the measurements that would confirm or refute each open claim.

Editorial Note. This document is formatted as a continuation of the Unified Dual-Helix Framework series and as an addition to *The Geometric Engine of Time* (Hoard 2026a). It supersedes the October 2026 technical report *Metrological Foundations* (Appendix A) and is written as a Stage 1 registered report: hypotheses, methods and decision rules are fixed before any new data are collected. Its constructions operate at laboratory and historical scale and are explicitly distinguished from the Engine's cosmological–temporal constructions and from the integer-factorization applications of the core SHD-CCP manuscripts. All quantitative claims are annotated with status tags: **[PROVEN]**, **[HEURISTIC]**, **[CONJECTURAL]**, **[OPEN]**, **[SUPERSEDED]**, refined by four tags defined under Epistemic Conventions and Scope.

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Epistemic Conventions and Scope

Every claim carries one of nine tags. Five are the series tags of the Geometric Engine of Time; the report's **[DEFINITION]** and **[EMPIRICAL]** are kept; and **[ESTABLISHED]** and **[COMPUTED]** are added so that cited physics and evaluated formulas are not labelled as the authors' proofs.

Table 1: Status tags used in this installment.

Tag	Meaning	Example in this proposal
[PROVEN]	A deductive result with a complete proof in this document. Series tag; exact arithmetic on stated inputs is tagged **[COMPUTED]** here.	$A_R = 16A^2$ (Theorem 4.2)
[DEFINITION]	A convention adopted by the proposal. It can be useful or not, but not true or false.	$C_0 := v/(16\, f_{cal})$

Tag	Meaning	Example in this proposal
[ESTABLISHED]	Standard physics or mathematics, cited and not re-proved here.	Laplace sound speed; Zwegers' completion of f(q)
[COMPUTED]	A number obtained by evaluating a stated formula. Reproducible from Appendix C.	$1 - \mathrm{sech}(u_\odot) = 2.60 \times 10^{-10}$
[EMPIRICAL]	A comparison with measured data, with source and uncertainty.	Royal cubit 52.375 ± 0.013 cm
[HEURISTIC]	A modelling choice motivated by analogy, not derived. Its parameters are free. Series tag.	Stellar sech regulator
[CONJECTURAL]	A claim that would need new evidence to be accepted. Series tag.	Acoustic calibration of ancient cubits
[OPEN]	A stated hypothesis awaiting its preregistered test. Series tag.	H1b, tested by Protocol B
[SUPERSEDED]	A report claim replaced by a corrected statement in this document. Series tag.	Scalar mock-theta law; "1/10,000 mile"

Rule for mixed claims. A computed value of a heuristic formula is tagged **[HEURISTIC / COMPUTED]**: the arithmetic is checked, but the physics is not thereby established. As in the Geometric Engine of Time, an exact result and its physical reading are tagged separately. The report tagged several mixed results PROVEN; Appendix A lists them.

What this proposal does not claim.

- It does not propose replacing the SI metre. C_0 is a derived, state-dependent engineering unit, traceable to the SI through the second and the measured state of the air.
- It does not claim that any ancient culture calibrated a cubit acoustically. It sets out how that claim would be tested and which results would refute it.
- It does not claim that the stellar, galactic, holonomy or mock-modular constructions describe physical effects. They remain **[HEURISTIC]** or **[CONJECTURAL]**, and Section 5 states whether each can be tested.
- It does not treat the Engine's tori, helices and gears as physical objects. It uses the Engine's exact arithmetic and identities, which hold whatever their physical reading; the Engine itself tags those readings **[HEURISTIC]**.
- It makes no health or biological claims about 40 Hz or 432 Hz sound.

Relation to the technical report. Section and proposition numbers in this proposal differ from the report's. Each result cites the report's number in brackets, and Appendix A maps every correction to its original location.

Relation to the series. This installment imports results from *The Geometric Engine of Time* (Hoard 2026a) and cites constructions of *The Torsional Markov Memory-Offload Pump* (Hoard 2026b) and *The Hyperbolic Sparsemax Pump* (Hoard 2026c). Section 1.6 lists each import and says whether the series construction is the most accurate way to calculate at that point; Appendix E verifies every imported result.

Notation. Inline mathematics is set in plain text (for example $C_0 = \lambda/16$); display equations are typeset.

1 Introduction

The cubit is among the oldest attested standards of length, yet its value differed by about a quarter between cultures. This proposal asks whether one acoustic rule can explain part of that spread, and builds a usable acoustic unit whatever the answer.

1.1 Metrological Background

Since 2019 the SI fixes the speed of light at 299,792,458 m/s and the caesium hyperfine frequency at 9,192,631,770 Hz. The metre is therefore independent of the medium in which it is used.

An acoustic wavelength is not. In an ideal gas, $\lambda = v/f$ with v proportional to $\sqrt{T}$, so a length tied to λ grows by about 0.17 % per kelvin near 20 °C. For an acoustic array this is a feature: element spacings expressed in wavelengths keep the array's phase geometry fixed when the air changes, while spacings in metres do not.

The technical report built on this idea as part of the SHD-CCP series, drawing its frequencies from The Geometric Engine of Time (Hoard 2026a). It defined the cubit as $\lambda/16$ at 40 Hz, noted that this reproduces values in the historical range, and added non-local factors intended to make the unit valid across astrophysical environments. This proposal keeps the first idea, tests the second, and quantifies the third.

1.2 Historical Cubit Values

Only values traceable to a measurement or a primary compilation are used in the analyses below.

Table 2: Sourced historical cubit values.

Unit	Length	Basis	Source
Egyptian royal cubit, Great Pyramid	52.375 ± 0.013 cm (20.620 ± 0.005 in)	weighted mean of eight determinations, Petrie §136	Petrie 1883, ch. 20
Egyptian royal cubit, King's Chamber	52.405 ± 0.010 cm (20.632 ± 0.004 in)	four wall lengths at the base, Petrie §52	Petrie 1883, ch. 7
Egyptian royal cubit, Old Kingdom monuments	52.10 to 52.60 cm (20.51 to 20.71 in)	seven pyramids and temples, Petrie §137	Petrie 1883, ch. 20
Egyptian cubit rods (14 rods)	52.35 to 52.92 cm	Lepsius 1865	Cubit
Nippur cubit	about 51.86 cm	copper-alloy bar, about 2650 BCE; its use as a standard is disputed	Cubit; Eckhard Unger
Late Assyrian standard, king's and big cubits	about 51.5, 55.0 and 56.6 cm	Khorsabad sculpture (Guralnick 1996)	Cubit
Roman cubitus	about 44.4 cm	1.5 Roman feet (Vitruvius)	Cubit
Greek pēchys	about 46 cm	standard forearm cubit	Cubit
Rabbinic cubit	48 to 57.6 cm; 54 cm for six handbreadths of 9 cm	Naeh; Karelitz	Cubit
Megalithic yard (not a cubit)	about 82.9 cm (2.72 ft)	Thom 1962; statistical support for a uniform unit is marginal	Megalithic yard

The report also cites a "Sacred/Long" cubit of 53.59 cm, a "Babylonian royal" cubit of 53.10 cm, a "Persian royal" cubit of 54.00 cm, a "Talmudic great" cubit of 55.10 cm and an "Assyrian short" cubit of 51.1 cm. None was found in the sources above, so they are excluded until primary sources are given.

1.3 Research Questions

1. **RQ1, realization.** Can C_0 be realized reproducibly, and with what uncertainty?
2. **RQ2, mathematics.** Are the report's mathematical results correct as stated, and what are their complete proofs?
3. **RQ3, non-local terms.** Do the stellar, galactic and holonomy factors have measurable consequences on Earth or elsewhere?
4. **RQ4, history.** Is there evidence that any historical cubit was set acoustically?
5. **RQ5, coincidences.** Do the report's numerical coincidences survive a look-elsewhere correction?

1.4 Hypotheses

- **H2, engineering (confirmatory).** A two-source array whose spacing tracks $C_0(T)$ keeps its interference-null angle fixed as temperature changes. An identical array with fixed spacing moves its null by the predicted $-0.035°$ per kelvin.
- **H1a, annual calibration (confirmatory).** A cubit equals v(T)/d, with d one of 640, 648 or 768 and T the mean annual temperature of its provenance, within 2 °C.
- **H1b, cold-season calibration (confirmatory, generated by the exploratory analysis in Section 7).** Egyptian royal cubits equal v(T)/648, the two-thirds fraction of a 432 Hz wavelength, with T the coldest-month mean temperature of their provenance.
- **H0, null.** Cubit lengths do not depend on the local speed of sound.
- **H3, design intent (exploratory).** Chambers built in whole cubits place a low-order acoustic mode near a preregistered target frequency more often than random rooms of the same size.

1.5 Contributions

1. A corrected definition of the local cubit using the Cramer (1993) real-air equation, with a GUM-style uncertainty budget (Section 3).
2. Complete proofs of all seven mathematical results, five of them corrected, and two strengthened results: an explicit formula for the Markov cycle current and a temperature-invariance lemma for cubit-built rooms (Section 4).
3. A quantitative comparison of the non-local factors with general relativity (Section 5).
4. A look-elsewhere analysis of the cosmological and numerical correspondences (Section 6).
5. A preliminary test of the historical hypothesis against climate data (Section 7).
6. Three preregistered protocols with decision rules and power analysis, and an 18-month plan (Sections 8 and 9).
7. A script that reproduces every number in this proposal (Appendix C).

8. A map of where the Geometric Engine of Time is the most accurate way to calculate (Section 1.6), three exact results that follow from it (Sections 4.1, 4.2 and 4.6), and a check of every result imported from it (Appendix E).

1.6 Relation to the Geometric Engine of Time

This installment uses the Geometric Engine of Time (Hoard 2026a) as a method wherever its exact constructions are the most accurate way to calculate, and cites it as a source elsewhere. Four places qualify: caesium-cycle bookkeeping, the quadrea identity, the prime wheel and the measurement window (Table 3). The Engine's geometric readings, with tori, helices and gears as physical carriers, are not used; the Engine tags them **[HEURISTIC]**, and so does this proposal.

Table 3: Where the Geometric Engine of Time and its companion papers enter this proposal.

Section	Series result	Role here	Most accurate method?
4.1 Caesium arithmetic	Hoard 2026a, §1.3 and Theorem 5.1	Caesium periods per pulse and per window	Yes. Integer arithmetic modulo 60 decides every closure exactly; the shortest closure is 2 pulses.
4.2 Quadrea identity	Hoard 2026a, Theorem 4.1 (TQF)	Area and collinearity from quadrances	Yes, for integer or rational anchors: exact, free of square roots, with no tolerance. Not in floating point for thin triangles (Remark 4.3).
4.6 Wheel and sieve	Hoard 2026a, §3.1 ($C_3 \times C_2$) and §5 (30 nodes)	Residue classes of prime candidates	Better than the report's base-6 wheel: the 30 nodes form $\mathbb{Z}/30$, whose 8 unit classes test 20 % fewer candidates. A mod-210 wheel saves more.
8.1 Protocol A	Hoard 2026a, Theorem 5.1 (0.5 s closure)	Coherent averaging window	Yes. 0.5 s is the shortest window holding whole periods of the caesium clock and of both series carriers, 40 Hz and 432 Hz, so lock-in estimates carry no leakage bias.
3.2 Calibration frequency	Hoard 2026a, §1.2 and §1.3	Origin of 40 Hz = (2/3) × 60 Hz and of the 432 Hz rules	Source only. Any SI-traceable 40 Hz source gives the same C_0.
4.3 Two-source envelope	Hoard 2026a, Proposition 3.1	The half-cycle offset π in π + π/6	Source only. The envelope is ordinary phasor addition.
4.4 Quaternion chains	Hoard 2026b; Hoard 2026a, §2.3	Gear chains; tier orientations	Exact routes exist: identity (1) for gear cycles, signed permutation matrices for tier orientations.
4.5 Cycle current	Hoard 2026b	Three-gear Markov chain	Source only. The tree-theorem formula is the exact method.
4.7 Mock theta function	Hoard 2026c	Proposed analogy	Source only. Zwegers' theory is the method.
6.3 Great Year	Hoard 2026a, §1.3	432 = 25,920/60	Source only. 25,920 is a convention; measured precession gives 25,772 years.

The remaining sections use standard physics and statistics, which are the accurate methods there.

2 Notation, Constants and Data Sources

All computations use the constants below; no value is tuned to produce a match. The report's symbol π for a stationary distribution is replaced by μ to avoid a clash with the circle constant.

2.1 Symbols

Table 4: Symbols.

Symbol	Meaning	Unit
t, T	air temperature; T = t + 273.15	°C, K
p	static pressure	Pa
h	relative humidity, as a fraction	1
x_w, x_c	mole fractions of water vapour and CO_2	1
γ, M, R	ratio of specific heats, molar mass, molar gas constant	1, kg/mol, J/(mol K)
v	speed of sound (the report's v_local)	m/s
f_cal, λ	calibration frequency; acoustic wavelength λ = v/f_cal	Hz, m
C_0	local acoustic cubit, λ/16	m
C_{40}, $C_{2/3}$, $C_{9/16}$	v/640, v/648, v/768 (Section 3.2)	m
Q_1, Q_2, Q_3; S	quadrances (squared distances); $S = Q_1 + Q_2 + Q_3$	m²
A_R, χ	quadrea $S^2 - 2(Q_1^2 + Q_2^2 + Q_3^2)$; coupling $\sqrt{A_R} / S$	m⁴, 1
r_s, R★, r★	Schwarzschild radius, stellar radius, orbital radius	m

Symbol	Meaning	Unit
u★, Γ★	stellar argument and factor sech u★	1
K, μ, J	transition matrix, stationary distribution, cycle current	1
q, H	unit quaternion; composite holonomy	1

2.2 Physical and Astronomical Constants

Table 5: Physical and astronomical constants.

Quantity	Value	Source
Speed of light c	299,792,458 m/s, exact	SI (2019)
Caesium frequency ν_Cs	9,192,631,770 Hz, exact	SI (2019)
Planck constant h	$6.626\,070\,15 \times 10^{-34}$ J s, exact	SI (2019)
Molar gas constant R	8.314 462 618 J/(mol K)	CODATA 2018
Newtonian constant G	$6.674\,30 \times 10^{-11}$ m³/(kg s²)	CODATA 2018
Planck length ℓ_P	$1.616\,255 \times 10^{-35}$ m	CODATA 2018
Bohr radius a_0	$5.291\,772\,109 \times 10^{-11}$ m	CODATA 2018
Dry-air molar mass M, ratio γ	0.028 964 7 kg/mol; 1.400	as in the report
Nominal solar radius R☉	6.957×10^8 m	IAU 2015 Resolution B3
Nominal solar mass parameter (GM)☉	$1.327\,124\,4 \times 10^{20}$ m³/s²	IAU 2015 Resolution B3
Nominal Earth radius and (GM)⊕	6.3781×10^6 m; $3.986\,004 \times 10^{14}$ m³/s²	IAU 2015 Resolution B3
Astronomical unit	149,597,870,700 m, exact	IAU 2012 Resolution B2
Distance to the Galactic Centre R_0	8.178 ± 0.013 (stat) ± 0.022 (sys) kpc	GRAVITY Collaboration 2019
R_0, alternative	8.275 ± 0.009 (stat) ± 0.033 (sys) kpc	GRAVITY Collaboration 2021, via MPE
Hubble constant H_0; Ω_Λ	67.66 ± 0.42 km/(s Mpc); 0.6889 ± 0.0056	Planck 2018, Table 2
General precession in longitude	5028.796195″ per Julian century	Capitaine et al. (2003), via Axial precession

2.3 Data Sources

- **Speed of sound in real air.** Cramer (1993), coefficients and auxiliary formulas as implemented in pyfar. Fitted range 0 to 30 °C, mole fraction of water vapour up to 0.06; stated uncertainty below 300 ppm.
- **Historical cubits.** The sources listed in Table 2.
- **Climate normals.** Cairo, daily mean 22.5 °C and January 14.4 °C, 1991 to 2020 (Cairo). Baghdad, Erbil, Rome, Athens, Jerusalem and Edinburgh from published station normals compiled in List of cities by average temperature.
- **Holocene temperature context.** Global mean surface temperature about 6500 years ago was 0.7 °C (5th to 95th percentile 0.3 to 1.8 °C) warmer than the 19th century (Kaufman et al. 2020).
- **King's Chamber geometry.** Petrie (1883), sections 51 to 55 (chapter 7).
- **Series results.** *The Geometric Engine of Time* (Hoard 2026a), as posted on Academia.edu; every result used here is re-derived in Appendix E.

2.4 Software

Python 3 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0 and mpmath 1.3.0. The script in Appendix C runs in about 25 seconds and stops at the first failed identity.

3 The Local Acoustic Cubit (Tier I)

Defined as one-sixteenth of the local acoustic wavelength at a fixed calibration frequency, C_0 can be realized at 20 °C with an expanded uncertainty of 0.22 mm (k = 2). It grows by about 0.98 mm per kelvin.

3.1 Speed of Sound in Air

Status: [ESTABLISHED]. Report Definition 2.1, with its pressure term corrected.

The ideal-gas (Laplace) speed of sound is:

$$v_{\mathrm{id}}(T) = \sqrt{\frac{\gamma R T}{M}}$$

With γ = 1.400 and M = 0.028 964 7 kg/mol this gives 331.32 m/s at 0 °C and 343.23 m/s at 20 °C **[COMPUTED]**. The report's tables instead scale 331.30 m/s by √(T/273.15), which differs by 0.006 %.

Real air departs from the ideal value through humidity, CO_2 and non-ideal behaviour. Cramer's (1993) equation covers these effects for 0 to 30 °C with an uncertainty below 300 ppm. At 20 °C, 101.325 kPa and 420 ppm CO_2 it gives 343.36 m/s for dry air and 343.98 m/s at 50 % relative humidity **[COMPUTED]**.

Pressure term, corrected. The report multiplies by $1 + \beta_P(p - p_0)$ with $\beta_P \approx 1.0 \times 10^{-7}$ Pa^{-1}. Cramer's equation gives $(1/v)\partial v/\partial p = -6.8 \times 10^{-10}$ Pa^{-1} at 0 °C, $+1.2 \times 10^{-9}$ Pa^{-1} at 20 °C and $+1.8 \times 10^{-9}$ Pa^{-1} at 30 °C **[COMPUTED]**. The report's coefficient is at least 55 times too large anywhere in 0–30 °C: at 83 kPa it predicts a −0.18 % change in the cubit, where real air gives −0.002 %.

The working definition of local sound speed is therefore:

$$v_{\text{local}} := v_{\text{Cr}}(t,\, p,\, x_w,\, x_c), \qquad 0 \le t \le 30\ ^{\circ}\text{C}$$

Outside 0 to 30 °C the ideal-gas formula with measured composition applies. It carries an extra model uncertainty that work package 1 will determine by direct measurement.

3.2 Definition of the Local Cubit

Definition (Local Acoustic Cubit). **[DEFINITION]** Report Definition 2.1; calibration frequencies from Hoard (2026a), §1.2 and §1.3.

$$\mathcal{C}_0(f_{\text{cal}};\, t, p, x_w, x_c) := \frac{\lambda}{16} = \frac{v_{\text{local}}(t, p, x_w, x_c)}{16\, f_{\text{cal}}}$$

The calibration frequency is realized from an SI-traceable reference, such as a GPS-disciplined oscillator. In the series it comes from the 60-pulse-per-second register of the Geometric Engine of Time (Hoard 2026a, §1.2): 40 Hz is two-thirds of that pulse rate, and the 432 of the 432 Hz rules is the Engine's macroscopic bound 25,920/60 (§1.3). These origins name the frequencies but add no constraint, because any traceable 40 Hz source gives the same C_0. The register matters instead for timing the measurement (Section 8.1).

The report uses four rules, named here as follows.

Table 6: The four acoustic rules.

Symbol	Rule	Length	Value at 20 °C, 50 % RH
C_{40}	λ/16 at 40 Hz	v/640	53.75 cm
$C_{2/3}$	two-thirds of λ at 432 Hz	v/648	53.08 cm
$C_{9/16}$	nine-sixteenths of λ at 432 Hz	v/768	44.79 cm
λ_{432}	full wavelength at 432 Hz	v/432	79.63 cm

C_{40} and $C_{2/3}$ differ by only 1.25 % (648/640), which equals about 7 K of air temperature. A cubit that fits one rule at temperature T fits the other near T + 7 K, so historical data alone can hardly separate them.

3.3 Voxel Partition and Phase

Proposition 3.1 (Voxel Partition and Phase). **[PROVEN]** Report Proposition 2.2.

Let a cube of edge $L = 4C_0$ be divided into 4 × 4 × 4 voxels of edge C_0. Then:

1. *one voxel occupies 1/64 of the volume, and its face 1/16 of a cross-section;*
2. *a displacement of one cubit along a plane wave advances the phase by π/8;*
3. *the cube edge equals λ/4, the distance from a pressure node to the adjacent antinode of a standing wave.*

Proof. (1) Volumes scale as the cube of the edge and areas as its square, so $C_0^3/(4C_0)^3 = 1/64$ and $C_0^2/(4C_0)^2 = 1/16$. (2) With $k = 2\pi/\lambda$ and $\Delta x = \lambda/16$, $k\Delta x = \pi/8$. (3) $4C_0 = \lambda/4$; in $\Psi(x) = 2A \sin(kx)\cos(\omega t)$ the node at $kx = 0$ and the antinode at $kx = \pi/2$ are λ/4 apart. ■

Editorial Note. The value λ/16 is forced once two design choices are made: a 4 × 4 × 4 grid, and an edge equal to a quarter wavelength. It is a **[DEFINITION]**: a design decision, not a property of nature.

3.4 Temperature-Invariant Room Modes

Lemma 3.2 (Temperature-Invariant Room Modes). **[PROVEN]** New in this proposal.

Let a rigid-walled rectangular enclosure have edges N_x C₀, N_y C₀ and N_z C₀, with C₀ realized in the air the enclosure contains. Then its mode frequencies are:

$$f_{lmn} = \frac{v}{2}\sqrt{\frac{l^2}{(N_x\mathcal{C}_0)^2} + \frac{m^2}{(N_y\mathcal{C}_0)^2} + \frac{n^2}{(N_z\mathcal{C}_0)^2}} = 8f_{\text{cal}}\sqrt{\frac{l^2}{N_x^2} + \frac{m^2}{N_y^2} + \frac{n^2}{N_z^2}}$$

They do not depend on temperature, pressure or composition.

Proof. Substitute C₀ = v/(16 f_cal); v cancels. ■

Consequences.

- The 4-cubit cube at f_cal = 40 Hz has its lowest rigid-wall mode at 80 Hz, twice the carrier.
- A built structure is fixed in metres, so the invariance holds only for structures whose dimensions track C₀(T), such as an actuated array. This is the premise of Protocol A (Section 8).
- A 20-cubit length built at the calibration temperature has axial modes at multiples of 16 Hz when f_cal = 40 Hz. The 40 Hz carrier is then 2.5 times the fundamental, not a mode, which corrects the report's Protocol I (Section 8).

3.5 Sensitivities and Uncertainty Budget

Status: [COMPUTED]. New in this proposal.

The budget below follows the GUM (JCGM 100:2008) for C_{40} at 20 °C, 50 % relative humidity and 101.325 kPa.

Table 7: GUM uncertainty budget for C_{40}.

Input	Standard uncertainty	Relative contribution to C_0
Air temperature	0.05 K	9.1×10^{-5}
Relative humidity	2 %	7.3×10^{-5}
CO_2 mole fraction	50 ppm	1.3×10^{-5}
Pressure	50 Pa	5.9×10^{-8}
Cramer equation	300 ppm bound, rectangular	1.73×10^{-4}
Calibration frequency	1×10^{-9}, relative	1.0×10^{-9}
Combined, k = 1		**2.09×10^{-4}**

The result is C_{40} = 53.748 cm with a standard uncertainty of 0.112 mm and an expanded uncertainty of 0.22 mm (k = 2). The cubit grows 0.98 mm per kelvin, and raising humidity from 0 to 100 % adds 1.96 mm at 20 °C.

The equation of state, not the sensors, dominates the budget. Realizing C_0 to better than 0.1 mm therefore requires measuring v directly, by time of flight or acoustic interferometry, instead of computing it from the state of the air. Work package 1 does this.

3.6 Recomputed Calibration Values

Status: [COMPUTED]. Report Charts 1 and 2.

The report's tables are reproduced exactly by v = 331.30 √(T/273.15), except the 20 °C rows. Those use the engineering value 343.00 m/s, which in the report's own model corresponds to 19.63 °C. A consistent 20 °C cubit is 53.63 cm for an ideal gas, 53.65 cm for dry real air and 53.75 cm at 50 % relative humidity, not 53.59 cm. Appendix B gives the full table.

4 Mathematical Foundations (Tier I)

All seven of the report's mathematical results are true once stated precisely. Five needed corrected statements or proofs, two are strengthened here, and every identity was also checked numerically (Appendix C). Proposition 4.1 and Theorems 4.2 and 4.9 use the Geometric Engine of Time, whose exact constructions are the most accurate method there (Section 1.6).

Table 8: Mathematical results, their sources and status.

Result in this proposal	Report reference	Series source	Status	Change
Proposition 3.1, voxel partition	Proposition 2.2	none	**[PROVEN]**	none

Result in this proposal	Report reference	Series source	Status	Change
Proposition 4.1, caesium arithmetic	Proposition 1.1	Hoard 2026a, §1.3 and Theorem 5.1	**[PROVEN]**	prime factorization completed; shortest and joint closures added
Theorem 4.2, quadrea and Weitzenböck bound	Theorem 3.4	Hoard 2026a, Theorem 4.1	**[PROVEN]**	rational form added; floating-point claim corrected; triadic equality case added
Proposition 4.5, two-source envelope	Proposition 3.3	Hoard 2026a, Proposition 3.1 (offset π)	**[PROVEN]**	second solution 11π/6 added
Proposition 4.6, quaternion delta chains	Proposition 5.1	Hoard 2026b	**[PROVEN]**	boundary convention fixed; logarithm branch corrected
Proposition 4.8, three-state cycle current	Lemma 5.2	Hoard 2026b	**[PROVEN]**	sign criterion proved; entropy production added
Theorem 4.9, wheel, mod 24 and sieve	Theorem 5.3	Hoard 2026a, §3.1 and §5	**[PROVEN]**	item (1) reclassified as a definition; 30-node wheel added
Section 4.7, mock theta completion	Section 4.1	Hoard 2026c	**[ESTABLISHED]**	transformation law corrected

4.1 Caesium Arithmetic and the Parity Closure

Proposition 4.1 (Caesium Arithmetic). **[PROVEN]** Report Proposition 1.1; series source Hoard (2026a), §1.3 and Theorem 5.1.

Let W_Σ = Σ (16 − 2n) over n = 0 to 4, so W_Σ = 60. Then ν_Cs/60 = 306,421,059/2, whose fractional part is 1/2. Thirty consecutive intervals of 1/60 s therefore contain exactly 4,596,315,885 caesium periods.

Proof. The prime factorization is ν_Cs = 2 × 3² × 5 × 7² × 47 × 44,351. Since 60 = 2² × 3 × 5:

$$\frac{\nu_{\mathrm{Cs}}}{60} = \frac{3 \cdot 7^2 \cdot 47 \cdot 44\,351}{2} = \frac{306\,421\,059}{2}$$

The numerator is odd, so the fractional part is 1/2, and thirty intervals give 15 × 306,421,059 = 4,596,315,885. ■

Correction. The report writes the factorization as 2 × 3² × 5 × 7 × 14,591,479. The product is right, but 14,591,479 = 7 × 47 × 44,351 is not prime. The report also defines one geometric cycle as 1 s with f_geo = 60 Hz; the intended reading, which Hoard (2026a, §1.2) states as f_geo = W_Σ/Δt_geo, is 60 pulses per 1 s cycle.

Remark (Shortest and Joint Closures). **[PROVEN]** The proposition is equivalent to ν_Cs ≡ 30 (mod 60), so k pulses hold a whole number of caesium periods exactly when k is even. The shortest closure is 2 pulses: 1/30 s, or 306,421,059 periods. The 30-node traversal of Hoard (2026a, Theorem 5.1) closes because its six channels contain the C_2 parity of Proposition 3.1 there; 15 nodes, C_3 across the five tiers, would hold 2,298,157,942.5 periods. Adding a carrier, the shortest window that also holds whole carrier periods is 6 pulses (0.1 s) at 40 Hz or 1 kHz and 10 pulses (1/6 s) at 432 Hz. For 40 Hz and 432 Hz together it is 30 pulses, exactly the Engine's 0.5 s traversal (Appendix C, section R). None of this constrains a physical length.

> **Editorial Note.** For timing bookkeeping the Engine's register is the most accurate method available: integer arithmetic modulo 60 decides every closure exactly, with no rounding. Its geometric reading, the 30-node traversal as a temporal engine, stays **[HEURISTIC]**, as in Hoard (2026a, Remark 5.2).

4.2 The Quadrea Identity and the Weitzenböck Bound

Theorem 4.2 (Quadrea Identity and Weitzenböck Bound). **[PROVEN]** Report Theorem 3.4; part (1) is the TQF Area-Stability Identity of Hoard (2026a), Theorem 4.1.

For three points with quadrances $Q_1, Q_2, Q_3 > 0$, let $S = Q_1 + Q_2 + Q_3$ and $A_R = S^2 - 2(Q_1^2 + Q_2^2 + Q_3^2)$. Then:

1. *$A_R = 16A^2 = 4Q_1Q_2 - (Q_1 + Q_2 - Q_3)^2$, where A is the triangle's area, so $A_R > 0$ exactly when the points are not collinear;*
2. *for non-collinear points, $0 < \chi = \sqrt{A_R}/S \le 1/\sqrt{3}$, with equality exactly when $Q_1 = Q_2 = Q_3$;*
3. *equivalently, without square roots, $0 < \chi^2 = A_R/S^2 \le 1/3$.*

Proof. (1) With side lengths a, b, c, so that $a^2 = Q_1$ and so on, Heron's formula gives $16A^2 = (a + b + c)(-a + b + c)(a - b + c)(a + b - c) = 2(a^2b^2 + b^2c^2 + c^2a^2) - (a^4 + b^4 + c^4)$. In quadrances this is $S^2 - 2\Sigma Q_i^2$, and expanding $4Q_1Q_2 - (Q_1 + Q_2 - Q_3)^2$ gives the same polynomial. (2) The identity $3\Sigma Q_i^2 - S^2 = (Q_1 - Q_2)^2 + (Q_2 - Q_3)^2 + (Q_3 - Q_1)^2 \ge 0$ gives $\Sigma Q_i^2 \ge S^2/3$, hence $A_R \le S^2/3$, with equality exactly when the three differences vanish. This is Weitzenböck's inequality $a^2 + b^2 + c^2 \ge 4\sqrt{3}\,A$ in quadrance form. (3) Square (2). ■

Remark 4.3 (Numerical Conditioning). **[COMPUTED]** The report presents the quadrance form as free of floating-point drift. For the observer–Sun–Galactic Centre triangle the opposite holds. With $R_0 = 8.178$ kpc and a 60° angle at the Sun, $A_R/S^2 \approx 2.6 \times 10^{-19}$, below the 2.2×10^{-16} resolution of IEEE double precision. Evaluated in double precision, A_R comes out as 3.37×10^{66} instead of the exact 4.28×10^{63}. Reliable values need exact integer or rational arithmetic, or $16A^2 = 4|u \times v|^2$ computed from two difference vectors u and v.

Remark 4.4 (Magnitude). **[COMPUTED]** For that triangle $\chi = (AU/R_0) \sin\theta$, where θ is the angle at the Sun between the directions to Earth and to the Galactic Centre. Hence $\chi \leq 5.9 \times 10^{-10}$, and $\chi = 5.13 \times 10^{-10}$ at $\theta = 60°$.

> **Editorial Note.** For anchors with integer or rational coordinates, evaluating A_R by the TQF identity of Hoard (2026a, Theorem 4.1) in exact arithmetic is the most accurate method: it is exact, needs no square roots, and decides collinearity without a tolerance. The Engine's triadic anchors, at phases 0°, 120° and 240° on a circle of radius R, give $Q_1 = Q_2 = Q_3 = 3R^2$, $A_R = 27R^4$ and $\chi^2 = 1/3$ exactly, the equality case of part (2). The gear's stability check therefore reduces to the exact test $3(Q_1^2 + Q_2^2 + Q_3^2) = S^2$. Its use as a predictor of signal quality stays **[HEURISTIC]**, as Hoard (2026a) states.

4.3 The Two-Source Interference Envelope

Proposition 4.5 (Two-Source Interference Envelope). **[PROVEN]** Report Proposition 3.3; the half-cycle offset π is the inverse-braid parity of Hoard (2026a), Proposition 3.1.

Two unit-amplitude sources at distances r_1 and r_2 from the observer emit the same frequency, the second with an extra phase $\pi + \pi/6$. Write $x = k(r_1 - r_2)$ and neglect the difference in 1/r spreading. The field is then a carrier with envelope $|\Pi| = 2|\sin((x - \pi/6)/2)|$, and:

- *$|\Pi| = 0$ exactly when $x \equiv \pi/6 \pmod{2\pi}$;*
- *$|\Pi| = 1$ exactly when $x \equiv \pi/2$ or $x \equiv 11\pi/6 \pmod{2\pi}$;*
- *$|\Pi|$ reaches its maximum 2 exactly when $x \equiv 7\pi/6 \pmod{2\pi}$.*

Proof. The field is Re[e^(−iΩt) (e^(ikr₁) + e^(i(kr₂ + π + π/6)))]. Its envelope is |1 + e^(iφ)| with φ = π + π/6 − x, and |1 + e^(iφ)| = 2|cos(φ/2)| = 2|sin((x − π/6)/2)|. The envelope equals 1 when sin((x − π/6)/2) = ±1/2, that is (x − π/6)/2 ≡ ±π/6 (mod π), giving x ≡ π/2 or x ≡ −π/6 ≡ 11π/6 (mod 2π). The zero and the maximum follow from sin = 0 and |sin| = 1. ■

Correction. The report gives only $x \equiv \pi/2$ for unit amplitude; $11\pi/6$ is the second solution. The equal-amplitude assumption holds in the far field or for equidistant sources and should be stated.

Application. For a source pair of spacing d seen in the far field at angle θ from its axis, $x = kd \cos\theta$. With the report's spacing $d = C_0 = \lambda/16$, $kd = \pi/8 < \pi/6$, so no direction has a null. The pattern's minimum lies on the axis at 0.131 of the pair's peak amplitude, a front-to-back ratio of 16.6 dB. Protocol A (Section 8) therefore uses $d = \lambda/4$, which puts a null at $\cos\theta = 1/3$, or $\theta = 70.5°$.

4.4 Quaternion Delta Chains

Proposition 4.6 (Quaternion Delta Chains). **[PROVEN]** Report Proposition 5.1; the gear chains are those of Hoard (2026b).

Let S^3 be the unit quaternions. Take outer-gear states $q_0, \ldots, q_{719}$ with the cyclic convention $q_{-1} := q_{719}$, which the report leaves undefined for the first block, and increments $\Delta q_k = q_k\, q_(k-1)^{-1}$.

1. **Block telescoping.** *For $m = 0, \ldots, 71$, the product $\Delta q_(10m+9)\, \Delta q_(10m+8) \cdots \Delta q_(10m)$ equals $q_(10m+9)\, q_(10m-1)^{-1}$. Consequently the product of all 720 increments, $\Delta q_{719} \cdots \Delta q_0$, equals 1 for every sequence of states.*
2. **Partition.** *The level sets $A_l = \{m : \lfloor 10m/72 \rfloor = l\}$ have sizes $|A_l| = \lceil 7.2(l+1) \rceil - \lceil 7.2l \rceil$, which is 8 for $l = 0$ and $l = 5$ and 7 otherwise, summing to 72.*
3. **Fractional lift.** *Every q in S^3 except -1 can be written uniquely as $q = \cos\varphi + \hat{u} \sin\varphi$ with φ in $[0, \pi)$ and $\hat{u}$ a unit pure quaternion ($\hat{u}$ arbitrary when $\varphi = 0$). With $\log q = \varphi\hat{u}$, the lift q^(1/N) = exp(φû/N) satisfies (q^(1/N))^N = q.*

Proof. (1) Each inner pair $q_j^{-1}\, q_j$ collapses by associativity, leaving the outer factors. Applied to the whole cycle this gives $q_{719}\, q_{-1}^{-1} = 1$. (2) The integer m lies in A_l exactly when $7.2l \leq m < 7.2(l+1)$, and an integer satisfies $m < y$ exactly when $m \leq \lceil y \rceil - 1$. The resulting intervals [0, 7], [8, 14], [15, 21], [22, 28], [29, 35], [36, 43], [44, 50], [51, 57], [58, 64] and [65, 71] are disjoint and cover 0 to 71. (3) Multiples of one pure quaternion û commute, so exp(φû/N)^N = exp(φû) = q. ■

Correction to (3). The report parametrizes q by a rotation angle θ in (−π, π) with q = cos(θ/2) + û sin(θ/2). That covers only the hemisphere Re q > 0, and it is not unique, because (θ, û) and (−θ, −û) give the same q. For example, q = −1/2 + (√3/2)i has rotation angle 4π/3. SLERP implementations usually replace q by −q when Re q < 0, which represents the same rotation; either convention works if it is stated.

Remark 4.7 (Holonomy). **[COMPUTED]** Two consequences matter for the report's Master Equation and its Step 5.

- Any product of unit quaternions has norm 1, so the factor $\|H\|^{-1}$ in the Master Equation is identically 1.
- When increments are differences of stored states on a cycle, the full-cycle holonomy is 1 by (1). The middle-gear product is also 1 in ascending order, but not in descending order ($|H - 1| = 1.97$ in the test). A non-trivial holonomy needs increments generated independently of the states, for example by integrating an angular velocity, a stated product order, and a measure such as the rotation angle $2 \arccos|\mathrm{Re}\, H|$.

Editorial Note. Rotations drawn from the cube group compose exactly. The tier orientations Rot_x(π/2) and Rot_y(π/2) of Hoard (2026a, §2.3) generate its 24 rotations, all signed permutation matrices, and adjacent tiers differ by a 120° rotation about a body diagonal (Appendix E); for such rotations integer arithmetic is the most accurate method. The gear increments of Hoard (2026b) are not of this kind. For them identity (1) gives the cycle product exactly, where a floating-point product of 720 quaternions reproduces it only to rounding error.

4.5 The Three-State Cycle Current

Proposition 4.8 (Three-State Cycle Current). **[PROVEN]** Report Lemma 5.2, strengthened; the three-gear chain is that of Hoard (2026b).

Let X_t be an irreducible Markov chain on the states a = 12, b = 10 and c = 8, with row-stochastic matrix K and stationary distribution μ (μK = μ). Let $J_{ij} = \mu_i K_{ij} - \mu_j K_{ji}$. Then:

1. *$J_{ab} = J_{bc} = J_{ca}$, written J;*
2. *$J = (\Pi_+ - \Pi_-)/Z$, with $\Pi_+ = K_{ab}K_{bc}K_{ca}$, $\Pi_- = K_{ac}K_{cb}K_{ba}$ and Z > 0 as below;*
3. *the sign of J is the sign of $\Pi_+ - \Pi_-$, so J = 0 exactly when $\Pi_+ = \Pi_-$ (Kolmogorov's criterion);*
4. *the entropy production per step, $\sigma = J \ln(\Pi_+/\Pi_-)$, is non-negative and vanishes exactly when J = 0.*

$$Z = T_a + T_b + T_c, \quad T_a = K_{ba}K_{ca} + K_{bc}K_{ca} + K_{cb}K_{ba}, \quad T_b = K_{ab}K_{cb} + K_{ac}K_{cb} + K_{ca}K_{ab}, \quad T_c = K_{ac}K_{bc} + K_{ab}K_{bc} + K_{ba}K_{ac}$$

Proof. (1) Stationarity at a gives $\mu_a(K_{ab} + K_{ac}) = \mu_b K_{ba} + \mu_c K_{ca}$, which is $J_{ab} = J_{ca}$; the same at b gives $J_{bc} = J_{ab}$. The diagonal entries cancel. (2) By the Markov chain tree theorem, $\mu_i = T_i/Z$, where T_i sums the weights of the spanning trees directed into state i. Then $\mu_a K_{ab} - \mu_b K_{ba} = (T_a K_{ab} - T_b K_{ba})/Z$, and four of the six products cancel in pairs, leaving $K_{ab}K_{bc}K_{ca} - K_{ac}K_{cb}K_{ba}$. (3) Z > 0 for an irreducible chain. (4) J and $\ln(\Pi_+/\Pi_-)$ have the same sign by (3). ■

Correction. The report attributes the strict sign condition to Kolmogorov's criterion, which only gives J = 0 exactly when $\Pi_+ = \Pi_-$. The strict inequality needs step (2), as in Hill (1977) and Schnakenberg (1976). The result was confirmed on 20,000 random chains.

4.6 The Base-6 Wheel, Mod 24 and the Bilinear Sieve

Theorem 4.9 (Base-6 Wheel, Mod 24 and Bilinear Sieve). **[PROVEN]** Report Theorem 5.3; by the Chinese remainder theorem the six residue classes mod 6 correspond to the six $C_3 \times C_2$ channels of Hoard (2026a), §3.1.

1. *Every prime p > 3 satisfies $p \equiv \pm 1 \pmod 6$.*
2. *Every integer n coprime to 6 satisfies $n^2 \equiv 1 \pmod{24}$.*
3. *For k ≥ 1 and s in {+1, −1}, the number n = 6k + s is composite exactly when $k = 6ab + s_2 a + s_1 b$ for some integers a, b ≥ 1 and signs s_1, s_2 with $s_1 s_2 = s$. The report's form $6ab + s_1 a + s_2 b$ is the same set after relabelling.*

Proof. (1) Residues 0, 2 and 4 (mod 6) are even, and residues 0 and 3 are multiples of 3, so a prime above 3 has residue 1 or 5 ≡ −1. (2) Since n is odd, n − 1 and n + 1 are consecutive even numbers and one is divisible by 4, so 8 divides (n − 1)(n + 1). One of n − 1, n, n + 1 is divisible by 3 and it is not n, so 3 divides $n^2 - 1$. Hence 24 divides $n^2 - 1$. (3) If k has the stated form, then $n = (6a + s_1)(6b + s_2)$ with both factors at least 5, so n is composite. Conversely, if n = uv with u, v > 1, every divisor of n is coprime to 6, so $u = 6a + s_1$ and $v = 6b + s_2$ with a, b ≥ 1. Expanding gives $uv = 6(6ab + s_2 a + s_1 b) + s_1 s_2$, so $s = s_1 s_2$ and k has the stated form. ■

Reclassification. Item (1) of the report's theorem, the "1.5-cube linear offset" with 1 + 1.5 × 4 = 7, is not a mathematical statement. It becomes a **[DEFINITION]**: the residue classes 5 and 1 (mod 6) are labelled "Step 5" and "Step 7".

Remark. Part (3) is a base-6 analogue of the sieve of Sundaram. It restates trial division over the classes 6k ± 1 and gives no faster primality test. It was checked up to 10^7: 664,579 primes, no mismatches.

Remark (The 30-Node Wheel). **[PROVEN]** Read the five tiers of Hoard (2026a) as C_5. The 30-node space $C_3 \times C_2 \times C_5$ is then cyclic of order 30 by the Chinese remainder theorem, and primes above 5 lie in its $\varphi(30) = 8$ unit classes: 1, 7, 11, 13, 17, 19, 23 and 29 (mod 30). In the Engine's labels these are the nodes with odd parity, triadic phase ±120° and tiers T_1 to T_4; that labelling is a **[DEFINITION]**. A wheel on these classes tests 8/30 of the integers against 1/3 for the base-6 wheel, 20 % fewer candidates: 2,666,666 against 3,333,333 up to 10^7 (Appendix C, section R). A mod-210 wheel, at 48/210, saves more, so the node space improves on the report's sieve without being optimal.

4.7 The Mock Theta Function f(q) and Its Completion

Status: [ESTABLISHED] mathematics; [CONJECTURAL] analogy. Report Section 4.1, whose scalar transformation law is **[SUPERSEDED]**; the analogy is with Hoard (2026c).

Ramanujan's third-order mock theta function is:

$$f(q) = \sum_{n \geq 0} \frac{q^{n^2}}{(1+q)^2(1+q^2)^2 \cdots (1+q^n)^2} = 1 + q - 2q^2 + 3q^3 - 3q^4 + 3q^5 - 5q^6 + 7q^7 - 6q^8 + \cdots$$

The coefficients were computed to q^{40} (Appendix C). Zwegers' theorem (2001, Theorem 3.6), as restated by Ortiz, Raum and Richter, concerns the vectors:

$$F^{+} = \begin{pmatrix} q^{-1/24} f(q) \\ 2q^{1/3}\,\omega(q^{1/2}) \\ 2q^{1/3}\,\omega(-q^{1/2}) \end{pmatrix}, \qquad G = \begin{pmatrix} -\sum_n (n+\frac{1}{6})\, q^{\frac{3}{2}(n+\frac{1}{6})^2} \\ \sum_n (-1)^n (n+\frac{1}{3})\, q^{\frac{3}{2}(n+\frac{1}{3})^2} \\ -\sum_n (n+\frac{1}{3})\, q^{\frac{3}{2}(n+\frac{1}{3})^2} \end{pmatrix}$$

Here ω is the companion third-order mock theta function. Adding to F^+ a fixed multiple of the non-holomorphic Eichler integral of G gives a vector-valued harmonic Maass form of weight 1/2 for the representation:

$$\rho(S) = \zeta_8^{-1} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad \rho(T) = \begin{pmatrix} \zeta_{24}^{-1} & 0 & 0 \\ 0 & 0 & \zeta_3 \\ 0 & \zeta_3 & 0 \end{pmatrix}$$

Correction. The report states a scalar law, f̂(−1/τ) = √(−iτ) f̂(τ). That is false and is **[SUPERSEDED]** by Zwegers' theorem above. The inversion τ → −1/τ exchanges the f-component with an ω-component, and the factor q^(−1/24) is required. The report's shadow, Σ(n + 1/6) e^(3πi(n + 1/6)²τ), is correct up to sign and normalization; its 1/6 comes from the character modulo 12 of the theta series, not from any π/6 phase gate.

> **Editorial Note.** Mapping the holomorphic part to the "Hyperbolic Sparsemax Pump" (Hoard 2026c) and the shadow to "8 offload bits" yields no derived consequence. It stays **[CONJECTURAL]** until it predicts a checkable property of the pump; otherwise it should be removed.

5 Non-Local Extensions and Their Testability (Tier II)

On Earth the stellar and galactic factors change C_0 by at most 2.6×10^{-10} and 1.2×10^{-9} $|\alpha_{\bowtie}|$, and the holonomy factor is identically 1. For $|\alpha_{\bowtie}| \leq 1$ that is at least five orders of magnitude below the 2.1×10^{-4} realization uncertainty, so no terrestrial measurement can test them.

5.1 A Locally Realized Unit Needs No Gravitational Correction

Status: [ESTABLISHED]. Replaces report Remark 3.2.

The Einstein equivalence principle includes local position invariance: in a small freely falling laboratory, non-gravitational physics is the same everywhere (Will 2014). C_0 is realized with a local clock and local air, so its local value carries no gravitational factor.

Gravity enters only when units realized at different potentials are compared, through the ratio of their redshift factors. To first order the relevant potentials are 9.87×10^{-9} for the Sun at 1 AU, 2.12×10^{-6} at the solar surface and 6.95×10^{-10} at Earth's surface ($GM/(c^2 r)$, **[COMPUTED]**).

This answers the question in the report's Remark 3.2. Terrestrial experiments need no astrophysical correction because of local position invariance, not because a regulator happens to be dormant.

5.2 The Stellar Factor

Status: [HEURISTIC / COMPUTED]. Report Section 3.1 and Proposition 3.1.

The report defines:

$$u_\star = \frac{r_s}{R_\star} \ln \frac{r_\star}{R_\star} + \eta_{\text{plasma}}, \qquad \Gamma_\star = \operatorname{sech} u_\star$$

Table 9: The stellar factor, as reported and recomputed.

Quantity	Report	Recomputed
Solar Schwarzschild radius r_s	2,953.25 m	2,953.25 m
r_s/R⊙	4.2449×10^{-6}	4.2450×10^{-6}
u⊙	2.281×10^{-5} in the statement, 2.2798×10^{-5} in the proof	2.2799×10^{-5}
$1 - \Gamma_\odot$	2.601×10^{-10}	2.599×10^{-10}
Shift of a 53.59 cm cubit	−0.139 nm, "less than two Bohr radii"	−0.139 nm, 2.63 Bohr radii
C_rel/C₀ on Earth	equal to 1 to "ten significant figures"	0.999 999 999 74, nine figures
Neutron star, r_s/R = 0.4, r = 10R	Γ = 0.687	Γ = 0.687; general relativity gives 0.980

The arithmetic is otherwise sound, but the construction has five problems.

1. **It is not derived from general relativity.** The logarithm of the tortoise coordinate sets the excess travel path of a radial signal near a mass: r_s ln(r/R) = 15.9 km from the Sun's surface to Earth, or 1.07×10^{-7} of the path. Dividing by R★ instead of by the path length is a free choice that inflates the argument by a factor of 214.
2. **It has the wrong order.** The hyperbolic secant makes the correction quadratic in r_s/R★, whereas the leading relativistic effects are linear. On Earth the sech term is 2.6×10^{-10} while the solar potential is 9.9×10^{-9}.
3. **It has the wrong strong-field behaviour.** At r = 10R around a neutron star with r_s/R = 0.4, it gives 0.687; the relativistic redshift factor √(1 − r_s/r) is 0.980.
4. **Its plasma term is unconstrained.** The report sets η_plasma = α_pl P_mag/P_gas and then neglects it. The solar-wind plasma beta at 1 AU is of order unity, so η is of order α_pl. Keeping $1 - \Gamma$ below 10^{-9} requires $|\alpha_{pl}| < 4.5 \times 10^{-5}$, a value nothing in the report fixes.
5. **One named ingredient is absent.** The abstract cites Brans–Dicke scalar-tensor coupling, which enters no equation.

Recommendation. Replace Γ★ by the ratio of relativistic redshift factors between realization site and comparison site, used only when units realized at different gravitational potentials must be compared. Otherwise remove it from the definition.

5.3 The Galactic Dual-Source Factor

Status: [HEURISTIC]. Report Section 3.2.

The report models the Galactic Centre as a coherent pair of sources with a fixed phase offset of π + π/6. Sagittarius A* is a single black hole of about four million solar masses (MPE). The report names no physical pair with that phase relation and gives no values for k_gal, Ω_gal or α_⋈.

- **Magnitude.** $|\Pi| \le 2$ and $\chi \le 5.9 \times 10^{-10}$ (Remark 4.4), so the term is at most 1.2×10^{-9} $|\alpha_{\bowtie}|$.
- **Sign.** The Master Equation uses Π itself, which changes sign at the frequency Ω_gal, whereas Proposition 4.5 bounds only the envelope |Π|. A length unit that oscillates in time needs an averaging rule, and none is given.
- **Testability.** Without a measurable k_gal and α_⋈ the term cannot be falsified.

Recommendation. Remove the factor from the definition, or specify a physical source, a coupling constant and a measurable consequence.

5.4 The Holonomy Factor

Status: [COMPUTED]. Report Section 4.3; the gears are those of Hoard (2026b).

Any product of unit quaternions has norm 1 (Remark 4.7), so $\|H\|^{-1}$ is identically 1. If a holonomy correction is intended, ε_hol must be defined as a function of the rotation angle of H, with a physical reason why a rotation should change a length.

5.5 The Master Equation, Restated

Status: [DEFINITION + HEURISTIC]. Report Master Equation.

With every parameter made explicit, the report's Master Equation reads:

$$\mathcal{C}_{\mathrm{rel}} = \mathcal{C}_0 \cdot \operatorname{sech}(u_\star) \cdot \left(1 + \alpha_{\bowtie} \Pi_{\bowtie} \chi\right) \cdot \left(1 - \epsilon_{\mathrm{hol}}\right)$$

On Earth this gives:

$$\left| \frac{\mathcal{C}_{\mathrm{rel}}}{\mathcal{C}_0} - 1 \right| \leq 2.6 \times 10^{-10} + 1.2 \times 10^{-9} \, |\alpha_{\bowtie}| + |\epsilon_{\mathrm{hol}}| + \text{higher-order terms}$$

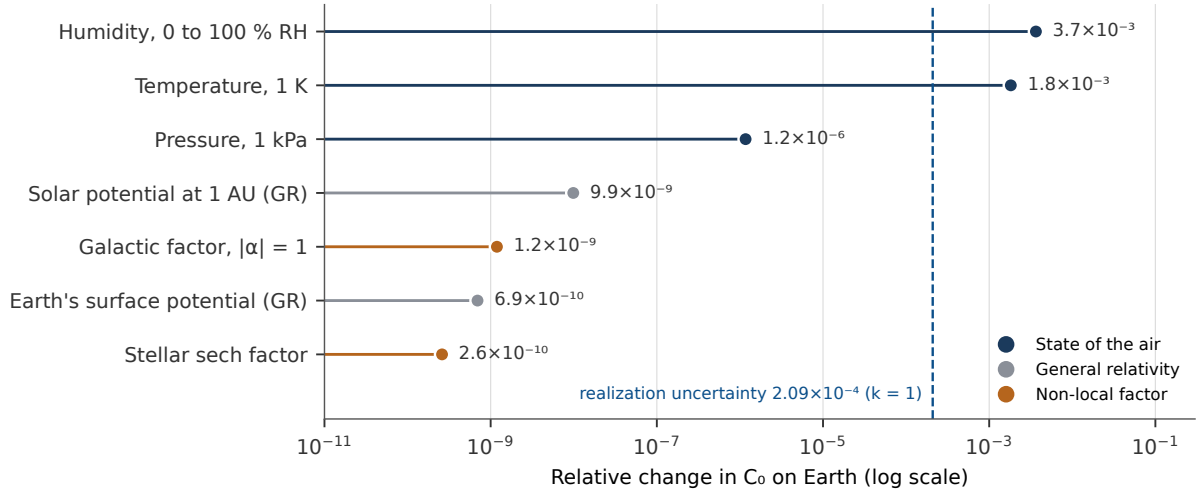


Figure 1: Every non-local term is at least 10^5 times below the realization uncertainty. Computed in Appendix C, sections E to G · relative effects on C_0 at 20 °C, galactic bound for $R_0 = 8.178$ kpc.

In temperature terms, the largest non-local term equals a change of less than 10^{-6} K, while the air itself moves C_0 by 1.8×10^{-3} per kelvin.

For metrology, C_0 is therefore the whole definition, and cross-site comparisons use relativistic redshift factors. The non-local factors may still be studied as mathematical models, but they are not part of a usable unit.

6 Cosmological and Numerical Correspondences (Tier III)

The 53.22 cm "cosmological cubit" rests on an unexplained choice of horizon and on a multiplier drawn from a family that would land an arbitrary anchor length in the cubit range 92 % of the time. It therefore carries no evidential weight, and the other numerical correspondences fail or are coincidences of digits.

6.1 The Geometric-Mean Length

Status: [COMPUTED]. Report Proposition 4.1.

The report forms ℓ_eq = √(ℓ_P L) from the Planck length and a cosmological horizon L, then multiplies by 10,368 = 144²/2. With Planck 2018 parameters, $\Lambda = 3H_0^2\Omega_\Lambda/c^2 = 1.1056 \times 10^{-52}$ m⁻². The result depends on which horizon is used:

Table 10: The geometric-mean length for four horizon choices.

Horizon	L (m)	ℓ_eq (μm)	ℓ_eq × 10,368 (cm)
Hubble radius, c/H_0	1.3672×10^{26}	47.01	48.74
Cosmic event horizon today	1.5686×10^{26}	50.35	52.20
Value used in the report	1.630×10^{26}	51.33	53.22
Asymptotic de Sitter radius, √(3/Λ)	1.6473×10^{26}	51.60	53.50

The report's 1.630×10^{26} m, described as "including matter density", matches none of the standard horizons. The cosmic event horizon today is 1.57×10^{26} m, or 16.6 billion light years.

The length ℓ_eq has a real physical relative. The dark-energy length (ħc/ρ_Λc²)^(1/4) = (8π/3)^(1/4) ℓ_eq = 87.8 μm, for an energy density of 3.32 keV/cm³. Torsion-balance tests of Newton's inverse-square law now reach separations of 52

μm without deviation (Lee et al. 2020). That is the context in which lengths of this size matter; it says nothing about cubits.

6.2 Look-Elsewhere Test for the Factor 10,368

Status: [COMPUTED]. New in this proposal.

Method. Treat the anchor length ℓ as log-uniform between 10^{-5} and 10^{-4} m. For a family of candidate multipliers N between 10^3 and 10^5, compute the fraction of anchors for which some N puts ℓN inside a target window.

- **Families.** All 228 integers of the form 2^i 3^j 5^k in range ("5-smooth"). The 89 integers in range of the form x, xy or xy/z built from the report's own constants: 2, 3, 4, 5, 6, 8, 10, 12, 16, 24, 30, 60, 64, 72, 144, 360, 432, 720 and 25,920 ("architecture products").
- **Targets.** T1, the whole report-model range of C_{40} for 0 to 37 °C (51.77 to 55.16 cm). T2, ±0.1 % around 53.17 cm. T3, ±0.1 % around any of the seven values in the report's Chart 1.

Table 11: Probability that a multiplier family places an arbitrary anchor length in each target window.

Multiplier family	T1, whole range	T2, 53.17 cm ± 0.1 %	T3, any Chart 1 value ± 0.1 %
5-smooth integers (228)	1.00	0.11	0.57
Architecture products (89)	0.92	0.04	0.24

Interpretation. The 0.1 % agreement with 53.17 cm was picked after the fact from seven tabulated values, so T3 is the relevant target. The chance probability is 0.24 to 0.57 before the choice of horizon is counted, and higher after it. For the report's anchor, 10,368 is the only architecture product inside T1. The 5-smooth numbers 10,125 and 10,240 also land there, at 51.97 and 52.56 cm.

6.3 Other Numerical Claims

Status: [COMPUTED]. Report Section 6 and Charts 1 and 2.

- **"1/10,000 of an Imperial mile (52.80 cm)".** False: 1/10,000 of a statute mile is 16.09 cm. 52.80 cm is a mile divided by 3,048, a coincidence of the digits 5280 (feet per mile) with 0.528 m.
- **"Platonic Great Year" of 25,920 years.** The measured general precession of 5028.796195″ per century gives 25,772 years (Axial precession). 25,920 is 0.58 % longer; it is a traditional round number, so 432 = 25,920/60, the macroscopic bound of Hoard (2026a, §1.3), is exact arithmetic on a convention. The report cites it as "Proposition 1.3" of that paper, which has no such proposition (Appendix A, item 31).
- **432 Hz.** It has no standing in acoustics standards, where concert pitch A_4 is 440 Hz (ISO 16:1975).
- **Megalithic yard as a 432 Hz wavelength.** That needs 46.0 °C dry air, against an Edinburgh annual mean of 9.3 °C. Statistical support for Thom's uniform yard is itself marginal (Megalithic yard).
- **4.96 cm micro-cubit and the 960 MHz antenna aperture.** λ/2π at 960 MHz is 4.970 cm. The acoustic length matches only near 20 °C, since it changes by 0.17 % per kelvin; no shared physics links the two.

6.4 Status of the Mock-Modular Analogy

Status: [CONJECTURAL]. Report Section 4.1; the analogy is with Hoard (2026c).

The mathematics is treated in Section 4.7. As a physical or architectural claim it remains an analogy without a derived consequence.

7 Historical Correspondence: Preliminary Analysis

With annual-mean temperatures, only 1 of 8 independent sourced cubits lies within 2 °C of the temperature its best-fitting acoustic rule requires, which is what chance predicts. A cold-season variant fits the Egyptian and Nippur cubits; it was found after looking at the data, so it is preregistered here as H1b rather than claimed.

7.1 Why Matching Rods to Temperatures Proves Nothing on Its Own

The speed of sound rises continuously with temperature, so each rule maps 0 to 37 °C onto a continuous band about 6.5 % wide. The rules C_{40}, $C_{2/3}$ and $C_{9/16}$ together cover 43.2 to 46.0 cm and 51.1 to 55.2 cm. By the intermediate value theorem every cubit in those bands "matches" some temperature. A match is informative only when the temperature is fixed independently, before the comparison.

7.2 Method

Status: [EMPIRICAL / COMPUTED]. New in this proposal.

1. For each sourced cubit L and rule divisor d (640, 648 or 768), the required temperature T* solves v_id(T*) = dL for dry air.
2. T* is compared with two climate anchors at the provenance: the annual mean and the coldest-month (January) mean, from modern station normals (Section 2.3).
3. A cubit counts as a hit when the rule closest to the anchor gives |T* − T_anchor| ≤ 2 °C. The tolerance reflects Holocene variability: global mean temperature 6500 years ago was only 0.7 °C (0.3 to 1.8 °C) above the 19th century (Kaufman et al. 2020).
4. The chance expectation is the probability that a length drawn uniformly from 44 to 57 cm scores a hit under the same rules and anchors. Tail probabilities use the Poisson-binomial distribution over the eight independent cubits; the King's Chamber value duplicates the Great Pyramid cubit and is shown but not counted.
5. Humidity is ignored in the baseline. At 50 % relative humidity, required temperatures fall by about 0.65 °C (13.25 °C dry versus 12.60 °C humid for the Great Pyramid cubit under $C_{2/3}$), which changes no conclusion.

7.3 Results

Table 12: Temperatures each acoustic rule requires for the sourced cubits.

Cubit	Length (cm)	Climate proxy	Annual / January mean (°C)	T* for C_{40} / $C_{2/3}$ / $C_{9/16}$ (°C)	Closest rule
Egyptian royal, Great Pyramid	52.375	Cairo	22.5 / 14.4	6.4 / 13.5 / 129.5	$C_{2/3}$
Egyptian royal, King's Chamber (duplicate)	52.405	Cairo	22.5 / 14.4	6.8 / 13.8 / 129.9	$C_{2/3}$
Nippur	51.86	Baghdad	22.8 / 9.7	1.0 / 7.9 / 121.6	$C_{2/3}$
Late Assyrian standard	51.5	Erbil	19.5 / 6.2	−2.8 / 4.0 / 116.1	$C_{2/3}$
Late Assyrian king's	55.0	Erbil	19.5 / 6.2	35.2 / 42.9 / 170.8	C_{40}
Late Assyrian big	56.6	Erbil	19.5 / 6.2	53.4 / 61.6 / 197.0	C_{40}
Roman cubitus	44.4	Rome	15.2 / 7.5	−72.2 / −67.2 / 16.2	$C_{9/16}$
Greek pēchys	46.2	Athens	19.2 / 10.2	−55.6 / −50.1 / 40.1	$C_{9/16}$
Rabbinic, six handbreadths	54.0	Jerusalem	18.3 / 9.8	24.1 / 31.5 / 154.8	C_{40}

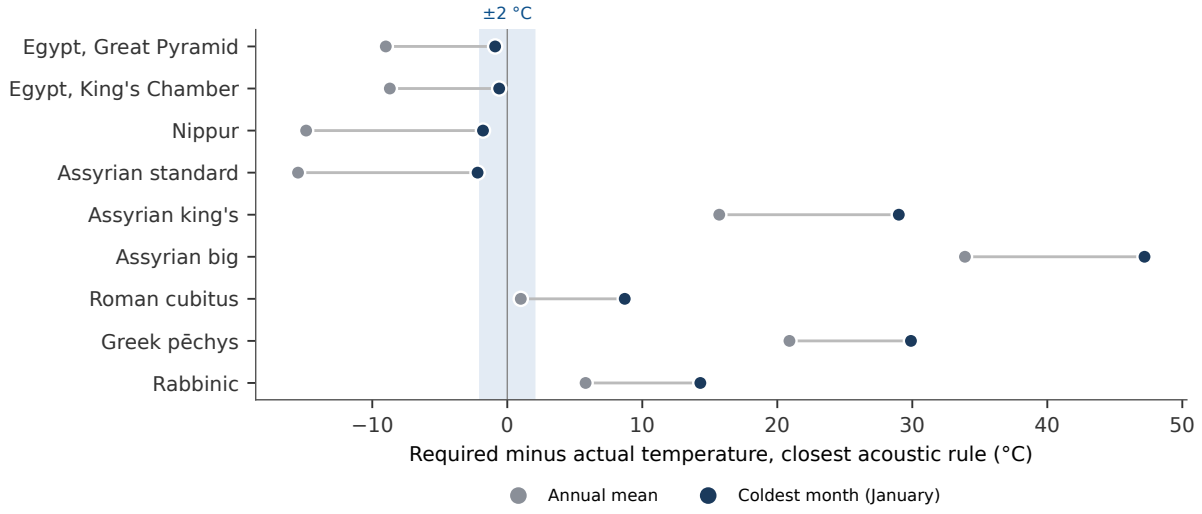


Figure 2: Cold-season temperatures fit Egypt and Nippur; annual ones fit only Rome. Computed in Appendix C, section N · cubit lengths from Section 1.2, climate normals from Section 2.3, dry ideal gas.

- **Annual-mean anchor (H1a).** One hit in eight: the Roman cubitus under $C_{9/16}$, 1.0 °C above Rome's mean. Chance predicts 0.64 hits, and P(at least 1) = 0.48.

- **Annual or coldest-month anchor.** Three hits in eight: the Egyptian royal cubit (−0.9 °C), Nippur (−1.8 °C) and Rome (+1.0 °C); the Late Assyrian standard cubit misses at −2.2 °C. Chance predicts 1.05 hits, and P(at least 3) = 0.076 before the choice of rules and anchors is counted.
- **No single rule fits.** The Egyptian and Mesopotamian hits use $C_{2/3}$ and the Roman hit uses $C_{9/16}$. Under $C_{2/3}$ the Roman cubit would need −67 °C.
- **Large misses remain.** The Assyrian king's and big cubits, the Greek pēchys and the rabbinic cubit miss by 6 to 34 °C under every rule and anchor.
- **The megalithic yard** needs 46.0 °C dry air as a full 432 Hz wavelength.

7.4 A Within-Climate Check on H1b

Petrie's seven Old Kingdom determinations around the Memphis necropolis share one climate. They range from 52.10 to 52.60 cm with a standard deviation of 1.78 mm (Petrie 1883, §137). Under H1b that scatter equals 2.0 K of calibration temperature, so it must come from year-to-year weather or workmanship. It also sets the residual scatter that Protocol B must assume.

7.5 Interpretation

H1a, calibration at the annual mean temperature, is not supported: its hit rate equals chance. The cold-season variant fits a coherent group, the Bronze Age Egyptian and Mesopotamian short royal cubits, with one rule and one anchor. It was identified only after the analyst chose among three rules and two anchors, so its p-value of 0.076 overstates the evidence. It becomes hypothesis H1b, to be tested on data not used here (Protocol B).

H1b is **[OPEN]** and makes a sharp prediction. Within the Egyptian royal-cubit family, cubit length should rise with the coldest-month mean temperature of the provenance by 0.91 mm per kelvin. A slope indistinguishable from zero refutes it.

7.6 Limitations

- Modern climate normals stand in for Bronze Age climate, and Cairo's urban heat island adds some warming.
- Site proxies are approximate: Baghdad for Nippur and Erbil for Khorsabad.
- Several cubit values come from secondary compilations, and the Nippur bar's status as a standard is disputed.
- Values the report cites without a source are excluded (Section 1.2).

8 Preregistered Validation Protocols

Three protocols test, in order, the engineering claim (H2), the historical claims (H1a and H1b against H0) and the archaeoacoustic design claim (H3). Each fixes its primary outcome, decision rule and sample size before data collection.

Table 13: Protocols, hypotheses and status.

Protocol	Hypotheses	Replaces the report's	Status
A, laboratory realization and tracking	H2	Protocol III	confirmatory
B, provenance-dated cubits and climate	H1a, H1b, H0	Protocol II	confirmatory
C, archaeoacoustic chambers	H3	Protocol I	exploratory

8.1 Protocol A: Laboratory Realization and Tracking (H2)

Why the report's Protocol III must change.

- Its testbed, 3.68 × 3.05 × 2.13 m (23.95 m³), has its lowest rigid-wall mode at 46.7 Hz. At 40 Hz, where λ = 8.60 m, the room sits below its first mode and behaves as a pressure chamber, not a free field. Its Schroeder frequency is 224 to 289 Hz for reverberation times of 0.3 to 0.5 s.
- With the report's spacing d = λ/16 the source pair has no null at all (Section 4.3).
- Its outcome, "maximum phase stability" of the quadrea A_R, is not defined.

Design.

- Stage A1, confirmatory, runs in an anechoic chamber at f_cal = 1 kHz. The tracking property does not depend on frequency, and 1 kHz lies well above typical anechoic cut-off frequencies.
- Two sources are driven with relative phase π + π/6 at spacing d = λ/4 = $4C_0$. A microphone arc at a radius of at least 10λ (3.4 m at 1 kHz) samples the pattern in 0.1° steps around the predicted null at 70.5°.

- Arm (i) keeps the spacing fixed at $4C_0$(20 °C). Arm (ii) tracks $4C_0$(T) with a linear actuator driven by a direct time-of-flight measurement of v.
- The chamber air is stepped from 15 to 25 °C in 1 K steps, with five replicate sweeps per step and arm. Temperature, humidity, CO_2 and pressure are logged throughout.
- Microphone amplitudes are estimated by lock-in detection over coherent windows of 0.5 s, the 30-pulse closure window of Hoard (2026a, Theorem 5.1). Each window holds whole numbers of carrier periods (500 at 1 kHz, 20 at 40 Hz), register pulses (30) and caesium periods (4,596,315,885), so the estimates carry no spectral-leakage bias; no shorter window does this for both series carriers, 40 Hz and 432 Hz (Section 4.1).

Prediction. With fixed spacing the null satisfies cos θ_null = λ/(12d), so it moves by −0.035° per kelvin, or −0.35° across the 10 K range. With tracked spacing it stays at 70.53°.

Primary outcome. The slope of the fitted null angle against measured temperature, for each arm. With a per-sweep null-angle uncertainty of 0.01°, the 55 sweeps per arm give a slope standard error of about 0.0004° per kelvin.

Decision rule. H2 is supported if both conditions hold. The tracked-arm slope lies within ±0.005° per kelvin of zero, by two one-sided tests at α = 0.05 each. The fixed-arm slope lies within ±20 % of −0.035° per kelvin. Otherwise the implementation has failed, and its cause is found before any 40 Hz work begins.

Stage A2, exploratory. In the testbed at 40 Hz, transfer functions from both sources to the eight-microphone array are recorded at each temperature and compared with a modal model of the room. No confirmatory claim is made.

Scope. Protocol A validates the sensors, the control loop and the realization of C_0. Its prediction follows directly from linear acoustics, so success says nothing about history or cosmology.

8.2 Protocol B: Provenance-Dated Cubits and Climate (H1a, H1b, H0)

Data.

- Physical lengths of graduated cubit rods and of well-dated architectural cubit determinations, each with provenance and date. Sources are museum collections of Egyptian rods, including those Lepsius compared, and published surveys of the kind in Petrie §137.
- Lengths are measured end to end at 20 °C with a calibrated coordinate-measuring machine or vernier, corrected for thermal expansion and documented damage. Two measurers work independently and blind to each provenance's climate.
- Climate covariates are the annual and coldest-month mean temperatures of each provenance and period: modern normals adjusted with regional Holocene reconstructions from the Temperature 12k database (Kaufman et al. 2020).
- The values used in Section 7 form a holdout and are excluded from the confirmatory test.
- Exclusions are fixed now: rods without provenance; broken rods whose length cannot be reconstructed; and model or votive rods more than 3 % from their family's median.

Primary analysis, H1b. Within the Egyptian royal-cubit family, an errors-in-variables (Deming) regression of measured length on the predicted length v(T_cold)/648. The error ratio comes from the measurement and climate uncertainties.

- H1b predicts a slope of 1, which is 0.91 mm per kelvin of coldest-month temperature.
- H0 predicts a slope of 0.
- H1b is supported if the 99 % confidence interval for the slope excludes 0 and includes 1. It is refuted if the interval excludes 1 and lies below 0.5. Any other outcome is reported as inconclusive.

Secondary analyses. H1a with annual means and all three rules, Holm-corrected for multiplicity. Cross-cultural comparisons are exploratory.

Sample size. The residual scatter within one climate is 1.8 mm (Section 7.4). Detecting the H1b slope at α = 0.01 with 80 % power then needs about 52, 18 or 11 independent provenance groups, if the coldest-month temperatures across sites have a standard deviation of 1, 2 or 3 K. If fewer than 18 groups are available by month 12, Protocol B reports an exploratory estimate only.

8.3 Protocol C: Archaeoacoustic Chambers (H3, Exploratory)

Why the report's Protocol I must change.

- Under the 40 Hz rule, a 20-cubit length has axial modes at multiples of 16 Hz (Lemma 3.2). 40 Hz is 2.5 times the fundamental, not a mode, so "integer mode-locking at 40 Hz" is not what the hypothesis predicts.

- The report's criterion, a mode within ±1 Hz of a multiple of 40 or 60 Hz or of 432/n, covers 37 % of the band from 16 to 130 Hz. At 20 °C the King's Chamber has 105 modes in that band; 26 meet the criterion, against 39 expected by chance. The criterion cannot discriminate.
- The chamber's temperature at construction is unknown, and today's temperature does not fix it.

For reference, the lowest rigid-wall modes of the King's Chamber at 20 °C and 50 % relative humidity, using Petrie's dimensions of 10.471 × 5.236 × 5.845 m, are 16.43 Hz (1,0,0), 29.43 Hz (0,0,1), 32.85 Hz (0,1,0 and 2,0,0), 33.70 Hz (1,0,1) and 36.73 Hz (1,1,0).

Design.

- **Hypothesis H3.** Chambers built in whole cubits place a low-order mode within ±δ of a preregistered target frequency f* at their measured equilibrium temperature more often than random rooms of the same volume class.
- **Measurements.** Impulse responses from swept-sine excitation, equilibrium air temperature and humidity, and as-built dimensions. δ equals the 95 % uncertainty of the measured mode frequencies.
- **Null model.** Mode statistics of rooms with random dimensions drawn from the same distribution of volumes and proportions, with expected mode counts from Weyl's formula. Wright (2007) used the same formula to show that resonances between 95 and 120 Hz in Neolithic chambers are expected from their size alone.
- **Decision rule.** One-sided p < 0.01 for each preregistered f*, Bonferroni-corrected across targets.
- **Status.** Exploratory unless at least 20 chambers are measured.

9 Research Plan

The work runs for 18 months in six work packages. Laboratory work and data assembly proceed in parallel, and no confirmatory data are collected before the Stage 1 protocol is accepted at month 2.

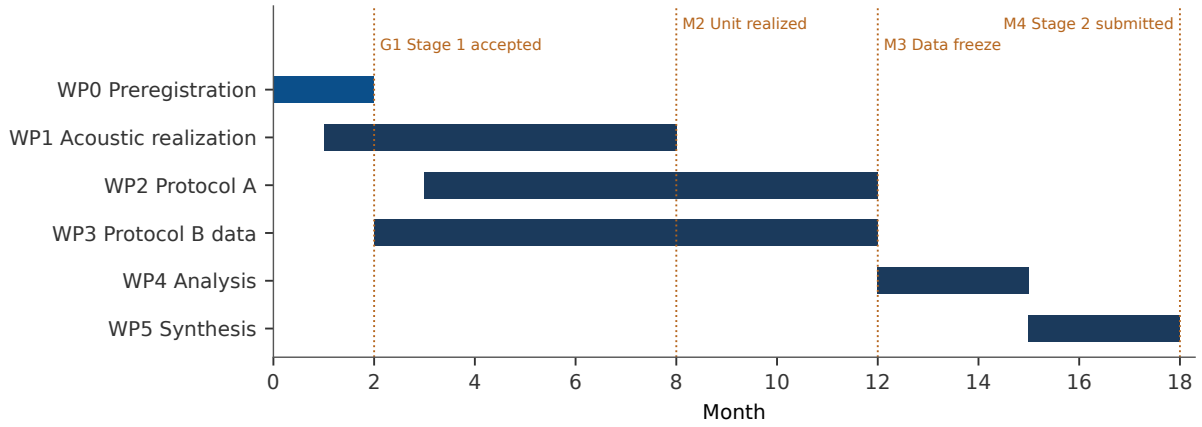


Figure 3: No confirmatory data are collected before Stage 1 acceptance at month 2. Research plan · six work packages and four gates over 18 months.

WP1 and WP3 start before the gate because they only build instruments and assemble data; every analysis that decides a hypothesis waits for G1.

9.1 Work Packages

1. **WP0, preregistration (months 0 to 2).** Submit Sections 7 and 8 as a Stage 1 protocol; deposit the verification code with a DOI.
2. **WP1, acoustic realization (months 1 to 8).** Measure v directly by time of flight against a GPS-disciplined clock, compare it with the Cramer equation, validate the uncertainty budget, and extend it beyond 0 to 30 °C.
3. **WP2, Protocol A (months 3 to 12).** Stage A1 in an anechoic chamber, then the exploratory Stage A2 in the testbed.
4. **WP3, Protocol B data (months 2 to 12).** Measure rods, document provenance and date, assemble climate covariates, and keep the measurers blind to climate.
5. **WP4, analysis (months 12 to 15).** Run the confirmatory Protocol B analyses and the exploratory Protocol C measurements.

6. **WP5, synthesis (months 15 to 18).** Write the Stage 2 report and a standalone note on the Tier I mathematics; release data and code.

9.2 Milestones and Decision Gates

Table 14: Milestones and decision gates.

Month	Milestone	Criterion
2	G1, Stage 1 accepted	No confirmatory data before this gate
8	M2, unit realized	C_0 realized with $U \leq 0.25$ mm ($k = 2$) by direct measurement
12	M3, data freeze	At least 18 provenance groups for Protocol B, or switch to an exploratory estimate
18	M4, Stage 2 submitted	Data and code public

9.3 Risks

Table 15: Risks and mitigations.

Risk	Likelihood	Impact	Mitigation
Fewer than 18 provenance groups	High	Protocol B underpowered	Add architectural determinations; otherwise report an exploratory estimate, as preregistered
Paleoclimate uncertainty of about 2 °C	Medium	Slope diluted towards zero	Errors-in-variables model; sensitivity analysis over plausible climates
No anechoic facility usable at 40 Hz	High	Stage A2 stays exploratory	Confirmatory Stage A1 runs at 1 kHz
Limited access to chambers	Medium	Protocol C delayed	Use published impulse responses; Protocol C is exploratory anyway
Analyst flexibility	Medium	False positives	Preregistration, blinded measurers, public code
Real-air equation outside 0 to 30 °C	Low	Larger uncertainty when hot	Direct measurement of v in WP1

9.4 Open Science

- Sections 7 and 8 are preregistered before any confirmatory data are collected, following the registered-report route.
- The verification script (Appendix C) and all analysis code are public with a DOI.
- Data are released under CC BY 4.0, except where heritage permits restrict publication of artefact measurements.

9.5 Ethics and Permits

- The work involves no human participants and no animals.
- Artefacts are measured only with institutional permission and by methods the holding museum approves.
- Staff exposure to low-frequency sound during 40 Hz work stays within occupational limits.

9.6 Budget Categories

Amounts are to be completed by the applicant.

- Personnel: an acoustician, an archaeometrist and a statistician, part time.
- Equipment: measurement microphones, a linear actuator stage, reference thermometry and hygrometry, and a GPS-disciplined frequency reference.
- Facility time in an anechoic chamber.
- Travel to collections for rod measurement.
- Open-access publication fees.

10 Status of Claims

Table 16 records the status of every principal claim in the series-standard vocabulary. Of the report's 20 principal claims, eight survive as definitions or proven results, five of them only after correction. Two rest on established results that the report misstated, four are computed values with heuristic or no physical meaning, and one is false. Of the remaining five, three stay heuristic or conjectural, the mode-locking test is redesigned as Protocol C, and the historical match is not supported as stated (H1a) and gives way to one preregistered hypothesis (H1b). The five rows marked "not in report" are new results of this proposal; three of them follow from the Geometric Engine of Time.

Table 16: Consolidated status of claims.

Claim	Report's tag	Status	Basis / Note	Testable on Earth?
Local acoustic cubit $C_0 = v/(16\, f_{cal})$	DEFINITION	**[DEFINITION]**	convention; realizable with U = 0.22 mm	yes, Protocol A and WP1
Sound speed with pressure coefficient 1.0×10^{-7} Pa^{-1}	PROVEN	**[ESTABLISHED]**, corrected	Cramer (1993); coefficient at least 55 times too large	yes
Voxel partition and quarter-wave edge	PROVEN	**[PROVEN]**	Proposition 3.1	not needed
Temperature-invariant modes of cubit-scaled rooms	not in report	**[PROVEN]**	Lemma 3.2	yes, Protocol A
432/40 = 10.8 and 25,920/60 = 432	PROVEN	**[COMPUTED]**	exact arithmetic on conventions; 432 is the macroscopic bound of Hoard (2026a, §1.3)	not needed
Caesium half-integer residual	PROVEN	**[PROVEN]**, proof corrected	Proposition 4.1; Hoard (2026a, Theorem 5.1)	not needed
Shortest and joint closure windows	not in report	**[PROVEN]**	Section 4.1; 0.5 s holds whole 40 Hz and 432 Hz periods	used in Protocol A
Quadrea identity and $\chi \le 1/\sqrt{3}$	PROVEN	**[PROVEN]**; floating-point claim corrected	Theorem 4.2; Hoard (2026a, Theorem 4.1)	not needed
Triadic anchors attain $\chi = 1/\sqrt{3}$	not in report	**[PROVEN]**	Section 4.2	not needed
Two-source interference envelope	PROVEN	**[PROVEN]**, second solution added	Proposition 4.5	yes, Protocol A
Quaternion telescoping, partition and lift	PROVEN	**[PROVEN]**, branch corrected	Proposition 4.6; Hoard (2026b)	not needed
Three-state Markov cycle current	PROVEN	**[PROVEN]**, strengthened	Proposition 4.8; Hoard (2026b)	not needed
Base-6 wheel, mod 24 and bilinear sieve	PROVEN	**[PROVEN]**; item (1) a **[DEFINITION]**	Theorem 4.9	not needed
30-node wheel with 20 % fewer candidates	not in report	**[PROVEN]**	Section 4.6; Hoard (2026a, §3.1 and §5)	not needed
Mock theta completion	CONJECTURAL	**[ESTABLISHED]** mathematics; scalar law **[SUPERSEDED]**; analogy **[CONJECTURAL]**	Zwegers (2001); Hoard (2026c)	no
Solar perturbation $1 - \operatorname{sech} u_\odot = 2.6 \times 10^{-10}$	PROVEN	**[HEURISTIC / COMPUTED]**	arithmetic sound; form disagrees with general relativity	no, far below uncertainty
Galactic dual-source factor	HEURISTIC	**[HEURISTIC]**, unfalsifiable as stated	no physical source; free parameters	no
Holonomy factor $\|H\|^{-1}$	PROVEN	**[COMPUTED]**: identically 1	Remark 4.7	not applicable
Match of acoustic rules to historical cubits	EMPIRICAL	**[EMPIRICAL]**; guaranteed by continuity; H1a not supported	Section 7	yes, Protocol B
Cold-season $C_{2/3}$ rule for Egyptian cubits (H1b)	not in report	**[OPEN]**, preregistered	Section 7	yes, Protocol B
Geometric mean ℓ_eq ≈ 51.33 µm	PROVEN ARITHMETIC	**[COMPUTED]**; 47.0 to 51.6 µm depending on horizon	Section 6.1	not applicable
ℓ_eq × 10,368 ≈ 53.22 cm	CONJECTURAL	**[CONJECTURAL]**, no evidential weight	look-elsewhere probability 0.24 to 1.00	no
"1/10,000 Imperial mile = 52.80 cm"	EMPIRICAL	**[SUPERSEDED]**: false	the true value is 16.09 cm	not applicable
Lorentz lift of the Centroid Monoid as generator of the integers	CONJECTURAL	**[CONJECTURAL]**	interpretation, no derived consequence	no
Mode-locking in ancient chambers	falsifiable test	criterion non-discriminating; redesigned as **[OPEN]** Protocol C	Section 8.3	exploratory, Protocol C

11 Conclusion

The acoustic cubit is a sound engineering unit, and its mathematics is correct once repaired. Its non-local extensions cannot be tested on Earth, and its historical claim is now one sharp hypothesis with a stated test. As in the Geometric

Engine of Time, the guarantees are asymmetric: what is proven stands, and what is heuristic is labelled as such.

What stands. $C_0 = v/(16\, f_{cal})$ can be realized to 0.22 mm (k = 2), and it keeps an array's phase geometry fixed as the air changes; Lemma 3.2 and Protocol A make that precise. All seven mathematical results hold with complete proofs, five of them after correction. The Geometric Engine of Time is the most accurate method at four points: caesium-cycle bookkeeping, the quadrea in exact arithmetic, the 30-node wheel and the 0.5 s coherent window.

What does not stand as physics. The stellar factor does not have the form general relativity gives, the galactic factor has no physical source, and the holonomy factor is identically 1. Even taken at face value, all three lie at least five orders of magnitude below the realization uncertainty. The 53.22 cm cosmological cubit is a look-elsewhere artefact, and the "1/10,000 mile" correspondence is an arithmetic error.

What remains open. Whether any culture set its cubit acoustically is unresolved. Calibration at the annual mean temperature fits no better than chance. Calibration at the coldest-month temperature with two-thirds of a 432 Hz wavelength fits the Egyptian and Nippur cubits, and Protocol B will test it on fresh data: a slope of 0.91 mm per kelvin confirms it, and a slope of zero refutes it.

A Errata and Audit of the Technical Report

The audit found 36 items, listed in the report's order. Only item 21 overturns a mathematical statement; the others correct proofs, numbers, tags, or physical and historical claims.

Table A1: Errata and audit of the technical report.

#	Location in the report	The report states	Finding	Resolution here
1	Abstract	Brans–Dicke coupling is part of the stellar carrier	It enters no equation	Remove or specify (5.2)
2	Proposition 1.1	"Prime factorization" $2 \times 3^2 \times 5 \times 7 \times 14{,}591{,}479$	$14{,}591{,}479 = 7 \times 47 \times 44{,}351$	$2 \times 3^2 \times 5 \times 7^2 \times 47 \times 44{,}351$ (4.1)
3	Proposition 1.1	One cycle $\Delta t_{geo} = 1$ s with $f_{geo} = 60$ Hz	A 1 s cycle has frequency 1 Hz	60 pulses per 1 s cycle (4.1)
4	Definition 2.1	Linear pressure term with $\beta_P \approx 1.0 \times 10^{-7}$ Pa^{-1}	Real air gives $\lvert\beta_P\rvert \le 1.8 \times 10^{-9}$ Pa^{-1} at 0 to 30 °C; humidity and CO_2 matter more	Cramer (1993) equation (3.1)
5	Table 3	Laplace and virial sound speed tagged PROVEN	Standard physics with a wrong coefficient	**[ESTABLISHED]**, corrected (10)
6	Table 3	432/40 = 10.8 and 25,920/60 = 432 tagged PROVEN	Exact arithmetic on conventions	**[COMPUTED]** (10)
7	Charts 1 and 2, 20 °C rows	v = 343.00 m/s at 20.00 °C	The report's model gives 343.21 m/s; 343.00 m/s corresponds to 19.63 °C	C_{40}(20 °C) = 53.63, 53.65 or 53.75 cm (3.6, Appendix B)
8	Chart 1	"1/10,000 Imperial mile (52.80 cm)"	1/10,000 of a mile is 16.09 cm	Removed (6.3)
9	Chart 1	Royal cubit placed at a "subterranean/vault mean" of 6.74 °C	No basis given; Cairo's annual mean is 22.5 °C	Tested against climate (7)
10	Charts 1 and 2	Sacred/Long 53.59, Babylonian royal 53.10, Persian royal 54.00, Talmudic great 55.10 and Assyrian short 51.1 cm	Not found in the sources checked; the Late Assyrian standard cubit is about 51.5 cm	Excluded until sourced (1.2)
11	Chart 2	Megalithic yard of 82.97 cm "comparable" at warm, humid conditions	Thom's value is 82.9 cm and needs 46.0 °C dry air	Rejected (6.3)
12	Proposition 3.1	$u_\odot = 2.281 \times 10^{-5}$ in the statement, 2.2798×10^{-5} in the proof	Inconsistent	2.2799×10^{-5} (5.2)
13	Proposition 3.1	ΔC is "less than two Bohr radii"	0.139 nm is 2.63 Bohr radii	Corrected (5.2)
14	Proposition 3.1	C_{rel} equals C_0 to "ten significant figures"	0.999 999 999 74 agrees to nine	Corrected (5.2)
15	Proposition 3.1	Tagged PROVEN	Arithmetic sound, but the model is not derived from general relativity and has the wrong order	**[HEURISTIC / COMPUTED]** (5.2)
16	Proposition 3.1	η_{plasma} set to zero	Solar-wind plasma beta is of order unity, so η is of order the free constant α_{pl}	Flagged (5.2)
17	Remark 3.2	The sech regulator explains why terrestrial experiments need no correction	Local position invariance explains it	Replaced (5.1)
18	Proposition 3.3	$\lvert\Pi\rvert = 1$ at $k\Delta r \equiv \pi/2$	Also at $11\pi/6$	Added (4.3)
19	Section 3.2	A dual source at the Galactic Centre	Sagittarius A* is a single black hole; no parameters are given	**[HEURISTIC]**, remove or specify (5.3)

#	Location in the report	The report states	Finding	Resolution here
20	Section 3.3	Quadrance triangulation avoids floating-point drift	$A_R/S^2 \approx 2.6 \times 10^{-19}$ for the astrophysical triangle; double precision fails	Exact arithmetic or a cross product (Remark 4.3)
21	Section 4.1	$\hat{f}(-1/\tau) = \sqrt{(-i\tau)}\,\hat{f}(\tau)$	False: the law is vector-valued and needs q^(−1/24)	Zwegers' theorem stated (4.7)
22	Section 4.2	$L_\Lambda \approx 1.63 \times 10^{26}$ m "including matter density"	Matches no standard horizon; the event horizon is 1.57×10^{26} m	Horizon table (6.1)
23	Proposition 4.1	53.22 cm agrees with the ISA cubit to 0.1 %	Arithmetic correct; results range from 48.7 to 53.5 cm with the horizon; chance probability 0.24 to 1.00	No evidential weight (6.2)
24	Section 4.3	$\|H\|_{Markov} = 1$ when "holonomically locked"	$\|H\| = 1$ for every product of unit quaternions	Factor removed (4.4, 5.4)
25	Section 4.3	$\Pi_{\bowtie}$ enters the Master Equation with its sign	It oscillates in time and no averaging rule is given	Flagged (5.3)
26	Proposition 5.1 (1)	ΔQ_0 uses q_{-1}	Undefined	Cyclic convention $q_{-1} := q_{719}$ (4.4)
27	Proposition 5.1 (3)	Unique polar form with θ in (−π, π)	Covers only Re q > 0 and is not unique	Half-angle in [0, π) (4.4)
28	Lemma 5.2	Sign of J follows from Kolmogorov's criterion	The criterion gives only J = 0 iff $\Pi_+ = \Pi_-$	Tree-theorem formula (4.5)
29	Theorem 5.3 (1)	"1.5-cube offset" with 1 + 1.5 × 4 = 7	Not a mathematical statement	**[DEFINITION]** of labels (4.6)
30	Section 6	Platonic Great Year of 25,920 years	Measured precession gives 25,772 years	Corrected (6.3)
31	Section 6, the 432 Hz standard	"Proposition 1.3 of The Geometric Engine of Time"	That paper has no Proposition 1.3; the bound 25,920/60 = 432 is in its §1.3	Cited as Hoard (2026a, §1.3) (3.2, 6.3)
32	Protocol I	Integer mode-locking at 40 Hz	40 Hz is not an axial mode under the 40 Hz rule; the criterion covers 37 % of the band	Protocol C (8.3)
33	Protocol III	Phase stability of the testbed at 40 Hz	The room is below its first mode at 46.7 Hz; d = λ/16 gives no null; the outcome is undefined	Protocol A (8.1)
34	Section 8, Step 5	Lie-group renormalization tagged PROVEN	No convergence proof; telescoped cyclic chains already give H = 1	Remark 4.7
35	Section 8, Step 1	Oscillator locked to the 30-node parity closure	Any SI-traceable 40 Hz source gives the same C_0	Simplified (3.2)
36	References	"Reid, 1997" cited in Protocol I	Missing from the reference list	Supply or remove

B Recomputed Calibration Tables

At 20 °C the consistent 40 Hz cubit is 53.63 cm for an ideal gas and 53.65 to 53.75 cm in real air, depending on humidity. Real-air columns use the Cramer (1993) equation at 101.325 kPa and 420 ppm CO_2; the 37 °C row extrapolates beyond its 0 to 30 °C fit.

Table B1: Speed of sound and cubit lengths from 0 to 37 °C.

t (°C)	v ideal (m/s)	v dry air (m/s)	v 50 % RH (m/s)	C_{40} ideal (cm)	C_{40} dry (cm)	C_{40} 50 % RH (cm)	$C_{2/3}$ dry (cm)	$C_{9/16}$ dry (cm)	λ_{432} dry (cm)
0	331.32	331.45	331.60	51.77	51.79	51.81	51.15	43.16	76.72
5	334.34	334.47	334.69	52.24	52.26	52.30	51.62	43.55	77.42
10	337.33	337.46	337.78	52.71	52.73	52.78	52.08	43.94	78.12
15	340.30	340.42	340.87	53.17	53.19	53.26	52.53	44.33	78.80
20	343.23	343.36	343.98	53.63	53.65	53.75	52.99	44.71	79.48
25	346.15	346.27	347.12	54.09	54.10	54.24	53.44	45.09	80.15
30	349.04	349.15	350.31	54.54	54.55	54.74	53.88	45.46	80.82
37, extrapolated	353.05	353.13	354.87	55.16	55.18	55.45	54.50	45.98	81.74

Derived lengths scale from these values: the 4 × 4 × 4 cube edge is $4C_{40}$, the report's π/6 boundary is $\lambda_{40}/12 = (4/3)C_{40}$, and its cymatic panel is $C_{40}/16$. Every value in the report's Charts 1 and 2 is reproduced by $v = 331.30\ \sqrt{(T/273.15)}$, except the 20 °C rows (Appendix A, item 7).

C Computational Verification

One script, verify_cubit_proposal.py, reproduces every number in this proposal, asserts every identity and re-derives every result imported from the Geometric Engine of Time; it runs in about 25 seconds and passes all checks. It accompanies this proposal and will be deposited with a DOI in WP0.

Running it. `python3 verify_cubit_proposal.py` runs every check; `--quick` limits the prime checks to 10^6. It needs numpy, scipy, sympy and mpmath, prints a report section by section, and writes the key values to results.json.

Table C1: Sections of the verification script.

Section	What it checks	Outcome
A	Constants and their sources	Values as in Section 2.2
B	Caesium factorization and the 60-pulse residual	Prime factorization corrected; 4,596,315,885 cycles confirmed
C	Ideal-gas, report-model and Cramer sound speeds; pressure coefficient	Report tables use $331.30\ \sqrt{(T/273.15)}$; $\lvert\beta_P\rvert \le 1.8 \times 10^{-9}$ Pa^{-1}
D	Every printed value in the report's Charts 1 and 2	Reproduced, except the 20 °C rows
E	GUM uncertainty budget	2.09×10^{-4} relative; U = 0.22 mm (k = 2)
F	Stellar factor against general relativity	$u_\odot = 2.2799 \times 10^{-5}$; shift of 2.63 Bohr radii
G	Quadrea identities, symbolically and on 200,000 random triangles; the astrophysical triangle in exact integers	Identities exact; double precision fails for the astrophysical triangle
H	Interference envelope on 400,001 points	Second solution 11π/6 confirmed
I	Quaternion telescoping on 72 blocks, partition sizes, 20,000 fractional lifts	All pass, including Re q < 0
J	20,000 random three-state chains	Equal currents, tree-theorem formula, entropy production ≥ 0
K	Primes up to 10^7	664,579 primes; wheel, mod 24 and sieve with 0 mismatches
L	Mock theta f(q) to q^{40}	1, 1, −2, 3, −3, 3, −5, 7, −6, …
M	Horizon choices and look-elsewhere coverage	Section 6 tables
N	Historical cubits against climate	1 of 8 and 3 of 8 hits; tail probabilities 0.48 and 0.076
O	Room modes of the King's Chamber and the testbed	105 modes from 16 to 130 Hz; testbed's first mode 46.7 Hz
P	Power analyses for Protocol B	Sample sizes in Section 8.2
Q	Mile, precession period, antenna aperture, ratio identities	Section 6.3
R	Results imported from the Geometric Engine of Time	All confirmed; shortest closure 2 pulses; joint 40 Hz and 432 Hz closure 30 pulses; tier clearances 10, 16, 22 and 28; cube-group orientations; 30-node wheel ratio 0.8; triadic $\chi^2 = 1/3$ (Appendix E)

Conventions. Random draws use a fixed seed (20261011), so every run gives identical output. Exact checks use Python integers, fractions and sympy, and floating-point checks state their tolerances in the code.

D Notation and Construction Catalogue

For completeness, every series symbol and construction this installment uses is listed below as the report or the Geometric Engine of Time defines it. Hoard (2026b, c) are cited for their constructions only and are not reviewed here; the results imported from Hoard (2026a) are verified in Appendix E.

Table D1: Notation and construction catalogue.

Symbol / Object	Meaning
SHD-CCP	The authors' computational architecture, within which the report's constructions are defined
$W_n = 16 - 2n$	Winding count of tier n, n = 0 to 4 (Hoard 2026a, §1.1)
$W_\Sigma = 60$	Total winding invariant, the sum of W_n over the five tiers (Hoard 2026a, Proposition 1.1)
τQ	Topological Quantum Pulse: one unit per full helical wrap (Hoard 2026a, §1.2)

Symbol / Object	Meaning
$\Delta t_{geo} = 1$ s; $f_{geo} = 60$ Hz	Definitional anchoring of one geometric cycle to the SI second, and the resulting pulse rate of 60 pulses per 1 s cycle (Hoard 2026a, §1.2)
$T_0 \dots T_4$	The five toroid tiers, T_0 innermost
$R_n = 5n^2 + 15n + 15$; $r_n = 2n + 4$	Major radii 15, 35, 65, 105 and 155; minor radii 4, 6, 8, 10 and 12 (Hoard 2026a, §2.1 and §2.2)
Rot(T_n)	$\mathrm{Rot}_x(\pi/2)$ for even n and $\mathrm{Rot}_y(\pi/2)$ for odd n (Hoard 2026a, §2.3)
$C_3 \times C_2$	Product symmetry of the six channels: three triadic phases 0°, 120° and −120° times two braid directions (Hoard 2026a, §3.1)
Parity map	$+1 \mapsto 0$ and $-1 \mapsto 0.5$, written π in Hoard (2026a, Proposition 3.1)
30 structural nodes	Six channels across five tiers; one traversal is 30 pulses, or 0.5 s (Hoard 2026a, §5)
$N_{pulse} = 153{,}210{,}529.5$; $\{N_{pulse}\} = 0.5$	Caesium periods per pulse, $\nu_{Cs}/60$, and its fractional part
$N_{total} = 4{,}596{,}315{,}885$	Caesium periods in one 30-node traversal, $\nu_{Cs} \times 0.5$ s
$T_{macro} = 432$	Great Year of 25,920 pulses divided by 60 (Hoard 2026a, §1.3)
Klein-wave merge boundary	$\lambda/12 = (4/3)C_{40}$, the distance over which the phase advances by π/6
Voxel telemetry cube, alignment mat	The 4 × 4 × 4 grid of cubit voxels, with edge λ/4
Cymatic panel	$C_{40}/16 = \lambda/256$
Quadrance, quadrea	A squared distance; $S^2 - 2\Sigma Q_i^2$, equal to sixteen times the squared area
Triple Quad Formula (TQF)	Wildberger's collinearity condition, quadrea = 0; Hoard (2026a, Theorem 4.1) uses the name for the area-stability identity $A_R = 16A^2$
χ_{TQF}	$\sqrt{A_R/S}$, the report's triangulation coupling ratio
Torsional Markov memory-offload pump	Three cyclic quaternion chains: a 12-point gear of 720 frames, a 10-point gear of 72 frames and an 8-point gear of 10 frames (Hoard 2026b)
Cycle current J	Net probability flow around the 12 → 10 → 8 → 12 cycle
Holonomy H	Product of the gears' net rotations
Hyperbolic Sparsemax Pump	A companion executable kernel (Hoard 2026c), referenced but not reviewed
Centroid Monoid, Lorentz lift	The report's seven-step narrative for the classes 6k ± 1; conjectural
Step 5, Step 7	Labels for the residue classes 5 and 1 (mod 6)
Dual-parity astrometric timing dipole, π/6 gate	The two-source model with relative phase π + π/6
ECM999 telemetry cube	Eight measurement microphones at the corners of the voxel cube
Gann Square of Nine, Master Square of 144	Grid layouts on the 4-cubit mat, with 19 × 19 and 12 × 12 tiles
Great Year	25,920 years in the report; the measured precession period is 25,772 years

E Results Imported from the Geometric Engine of Time

Every result this installment takes from Hoard (2026a) was re-derived in section R of the verification script. All hold as stated. Three passages need corrected wording, and five results can be sharpened.

Table E1: Results of Hoard (2026a) checked for this installment.

GET result	Statement	Check	Note	Used in
Proposition 1.1, winding invariant	$W_\Sigma = \Sigma(16 - 2n) = 60$	confirmed	A count fixed by the chosen winding law	4.1
§1.2, register	$\Delta t_{geo} = 1$ s, so $f_{geo} = W_\Sigma/\Delta t_{geo} = 60$ Hz	confirmed	f_{geo} is a pulse rate; the cycle frequency is 1 Hz. The Engine tags the anchoring **[HEURISTIC]**	3.2, 4.1
§1.3, macroscopic bound	25,920/60 = 432	confirmed	25,920 is a convention; measured precession gives 25,772 years. Tagged **[COMPUTED]** here	3.2, 6.3
§1.3, microscopic bound	$\nu_{Cs}/60 = 153{,}210{,}529.5$	confirmed	Equivalent to $\nu_{Cs} \equiv 30 \pmod{60}$	4.1
Proposition 2.1, major radii	$R_n = 5n^2 + 15n + 15$	confirmed; wording	The proof calls the first differences 20, 30, 40 and 50 "second differences"; the second differences are 10	not used

GET result	Statement	Check	Note	Used in
Proposition 2.2, minor radii	r_n = 2n + 4	confirmed	none	not used
§2.3, alternating embedding	Rot_x(π/2) for even tiers, Rot_y(π/2) for odd	sharpened	Clearances R_(n+1) − R_n − r_n − r_(n+1) of 10, 16, 22 and 28 keep concentric tiers apart in any orientation, so the embedding is not needed for separation. Adjacent tiers differ by a 120° rotation about a body diagonal; all orientations lie in the 24-element rotation group of the cube	4.4
Proposition 3.1, parity map	π(+1) = 0, π(−1) = 0.5	sharpened; wording	An injective homomorphism $C_2 \to \mathbb{Q}/\mathbb{Z}$, with φ_p = π((−1)^p). The symbol π also denotes the circle constant in §2.3	4.1, 4.3
Proposition 3.2, angular velocity	$\omega_0/\omega_4 = 16/8 = 2$	confirmed	"Acceleration" names a step increase of ω between tiers, not a change in time	not used
Theorem 4.1, TQF identity	$A_R = 16A^2$	confirmed; sharpened	Exact only in exact arithmetic (Remark 4.3); triadic anchors attain χ = 1/√3	4.2
Theorem 5.1, parity closure	30 pulses = 0.5 s hold 4,596,315,885 caesium periods	sharpened	Every even pulse count closes, the shortest being 2; 0.5 s is the shortest joint closure with the 40 Hz and 432 Hz carriers	4.1, 8.1
§5, 30 structural nodes	six channels × five tiers	sharpened	$C_3 \times C_2 \times C_5 \cong \mathbb{Z}/30$, with 8 prime-admissible nodes	4.6
Abstract	"the quadratic major radius law W_n = 16 − 2n"	wording	W_n is the linear winding law; the quadratic law is R_n	none

Editorial Note. None of these checks touches the Engine's geometric readings, which Hoard (2026a) itself tags **[HEURISTIC]**: the anchoring Δt_geo = 1 s, the alternating embedding, the dual helix as a parity carrier, the TQF gears as stability monitors and the 30-node traversal as a temporal engine.

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Links are given for every source opened while preparing this proposal; the remaining entries are standard works and the report's own citations.

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Climate normals. Cairo (Cairo International Airport, 1991 to 2020) and List of cities by average temperature for Baghdad, Erbil, Rome, Athens, Jerusalem and Edinburgh, each with its station source.

Submission details. Corresponding author: [name, affiliation, ORCID and email to be completed before submission]. Keywords: Geometric Engine of Time; SHD-CCP; acoustic metrology; speed of sound in air; ancient metrology; cubit; rational trigonometry; Weitzenböck inequality; non-equilibrium Markov chains; mock theta functions; local position invariance; look-elsewhere effect; registered report.